Cost of living

By: Mitchell Tope

Introduction

Data set Cost of Living US

The dataset has over 30,000 rows of information

Information incluedes:

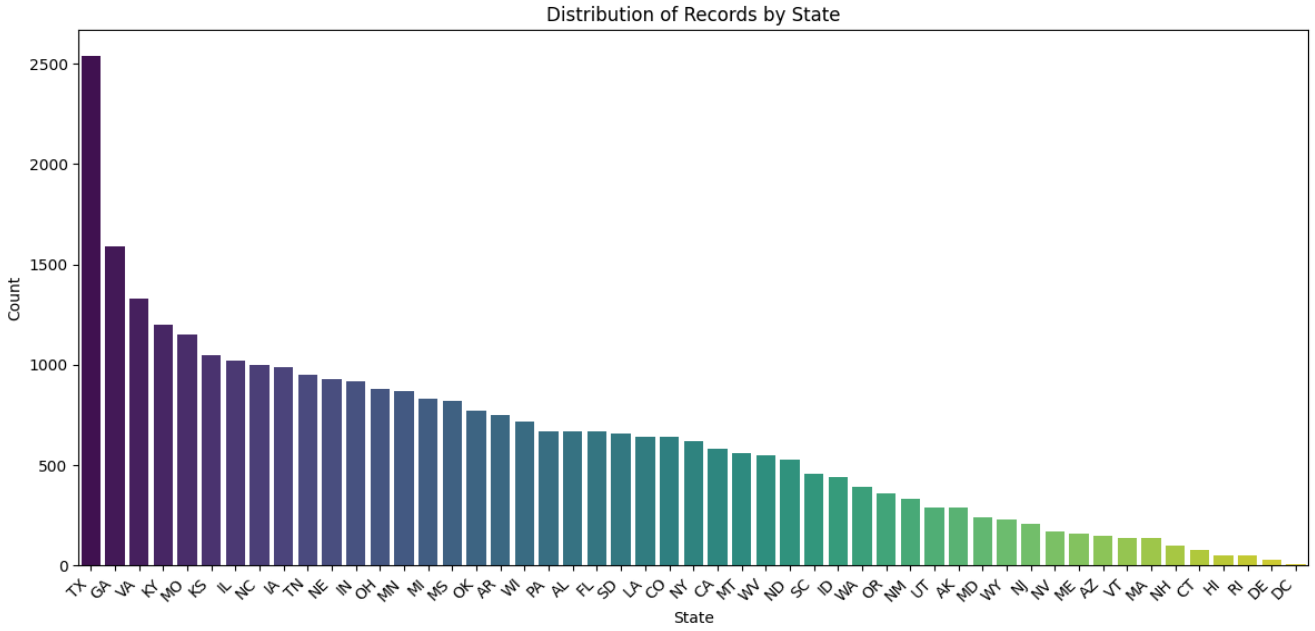
Family information	Locations	Income	Costs

Variables

- 1. case_id
- 2. state
- 3. isMetro
- 4. areaname
- 5. county
- 6. family_member_count
- 7. housing_cost
- 8. food_cost
- 9. transportation_cost
- 10. healthcare_cost
- 11. other_necessities_cost
- 12. childcare_cost
- 13. taxes
- 14. total_cost
- 15. median_family_income

State

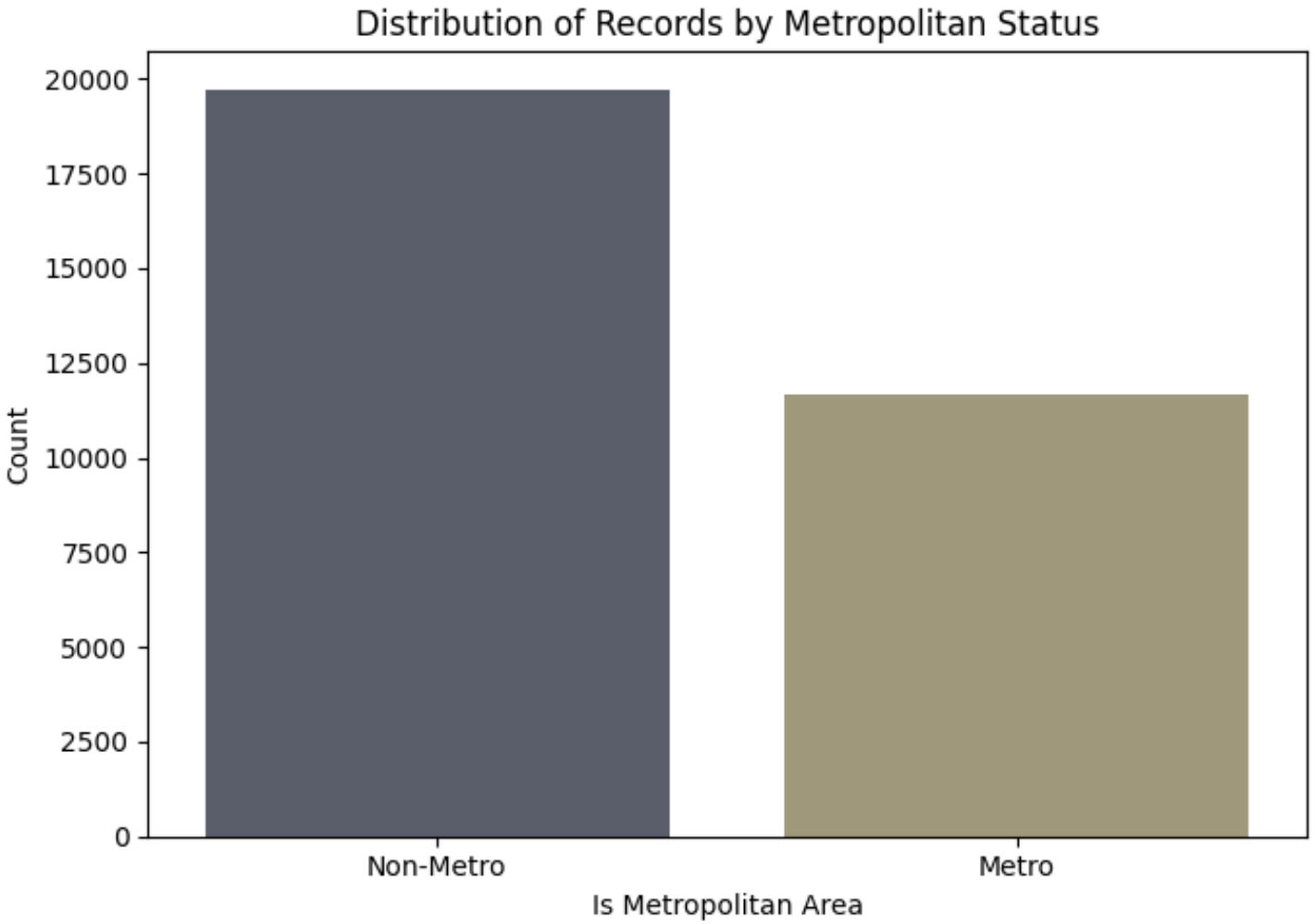
- Most comes from Texas
- Least comes from Washington Dc



IsMetro

In this variable we see how much of the variables are metropolitan.

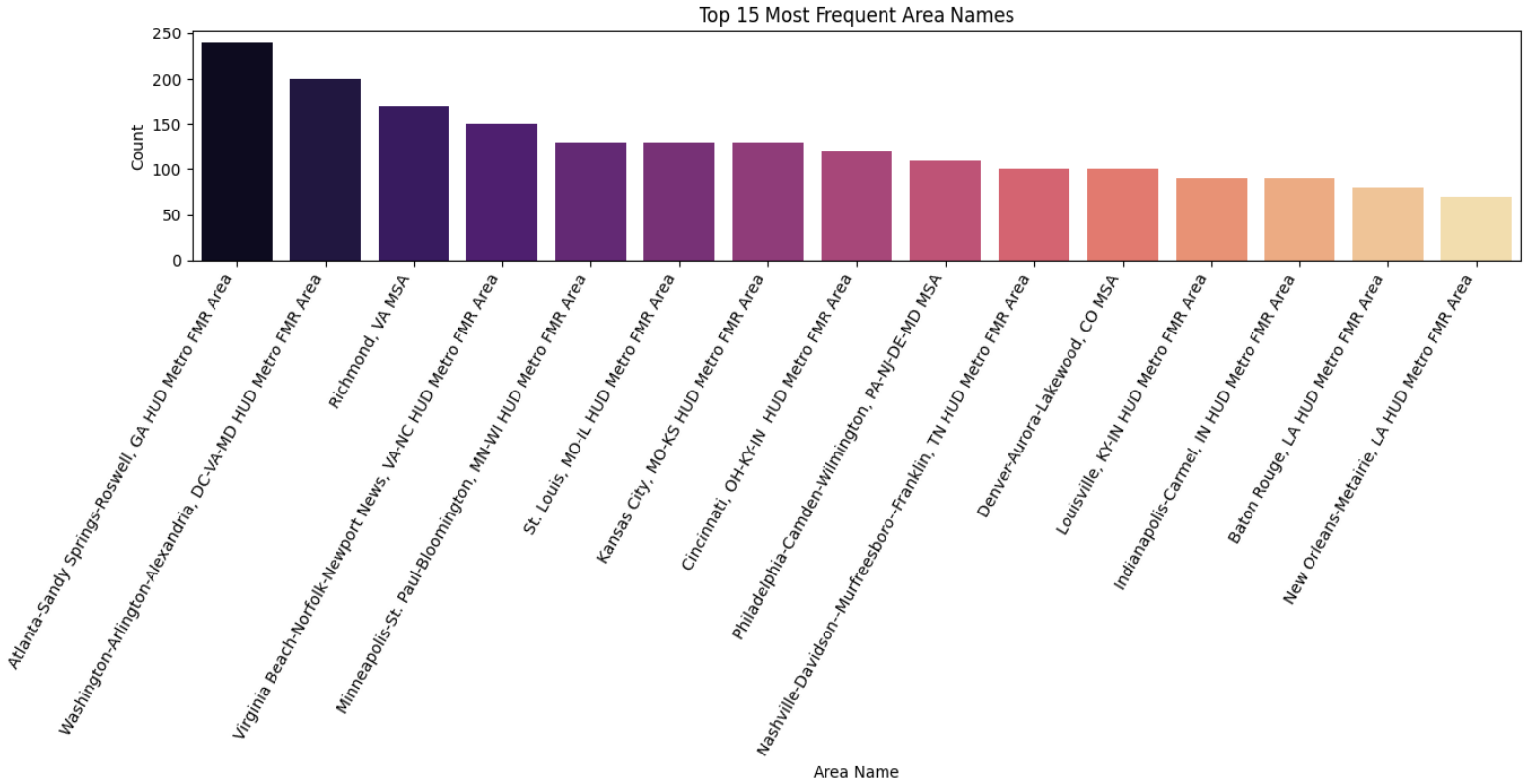
- Majority of the data is Non-Metro



Area name

In the graph, we see the 15 most used area names within the data.

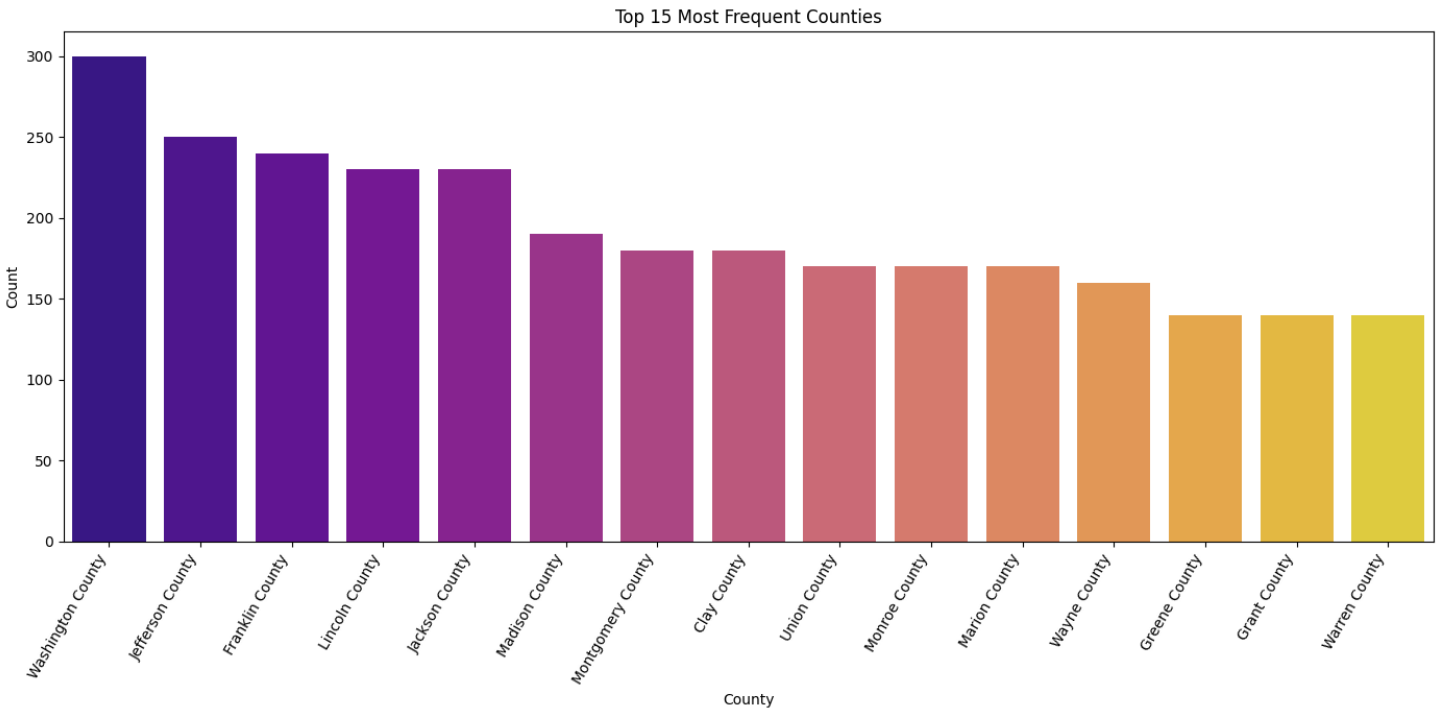
- Highest one is: Atlanta-Sandy Spring



County

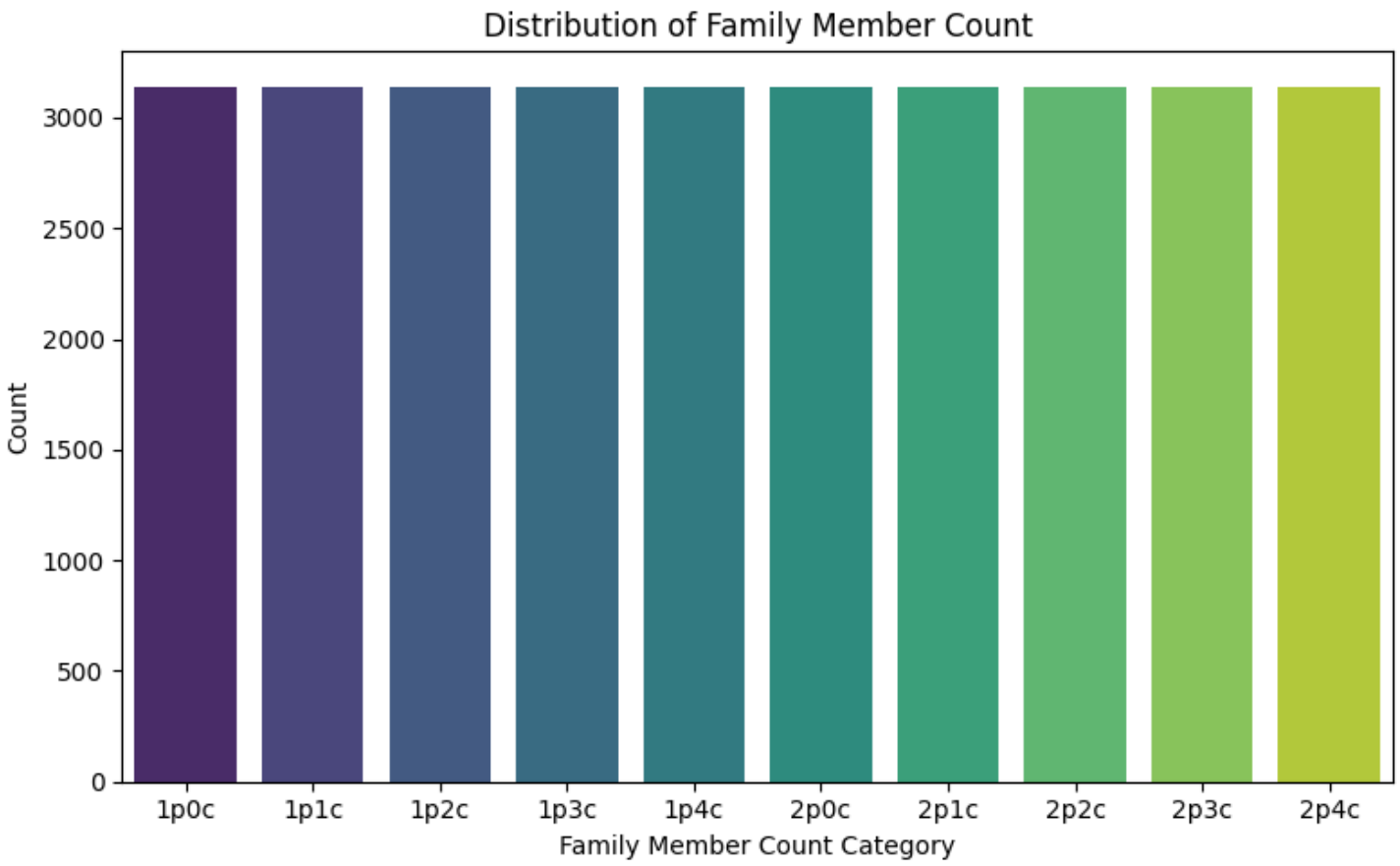
In this graph, we see the top 15 used counties

- Highest used county is Washington County (300)



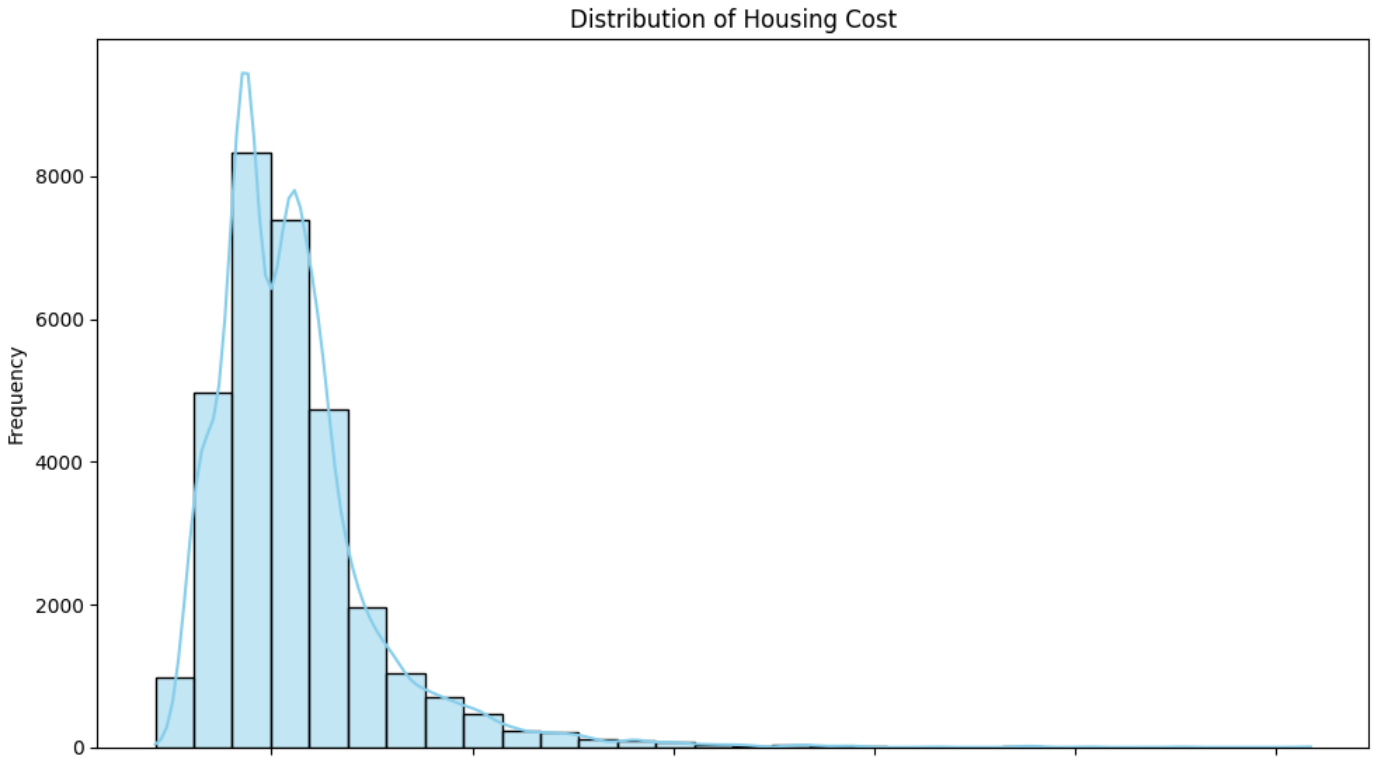
Family Member Count

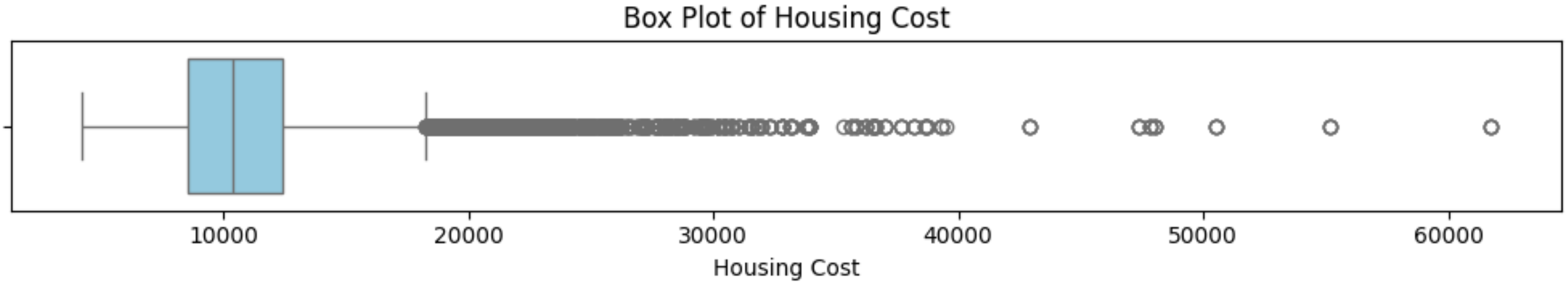
When it comes to family membern count we ccan see that the data appears to be equal.



Housing Cost

- Range: $57,526.28
  - Min: $4,209.31
  - Max: $61,735.59
- Average: $11,073.36

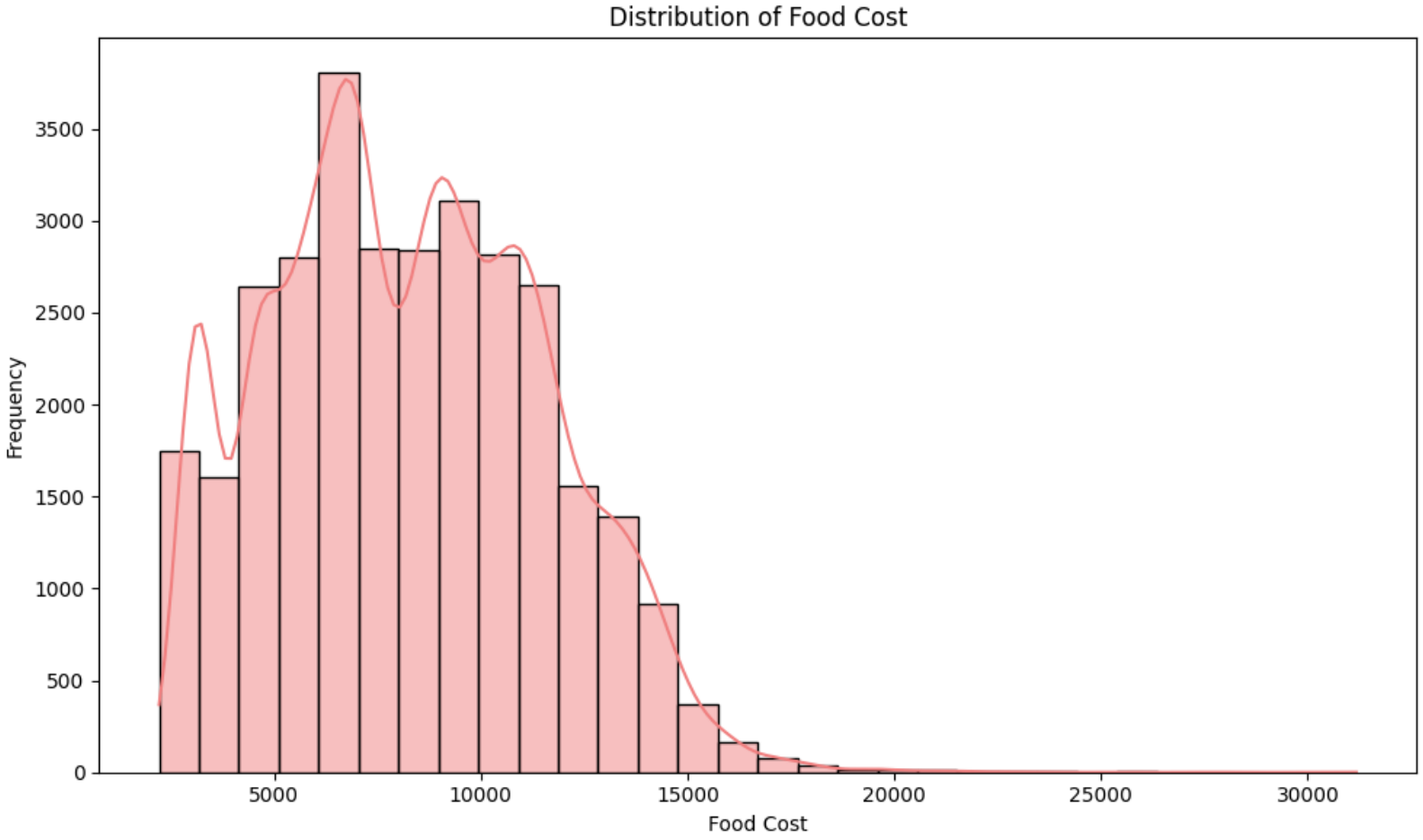


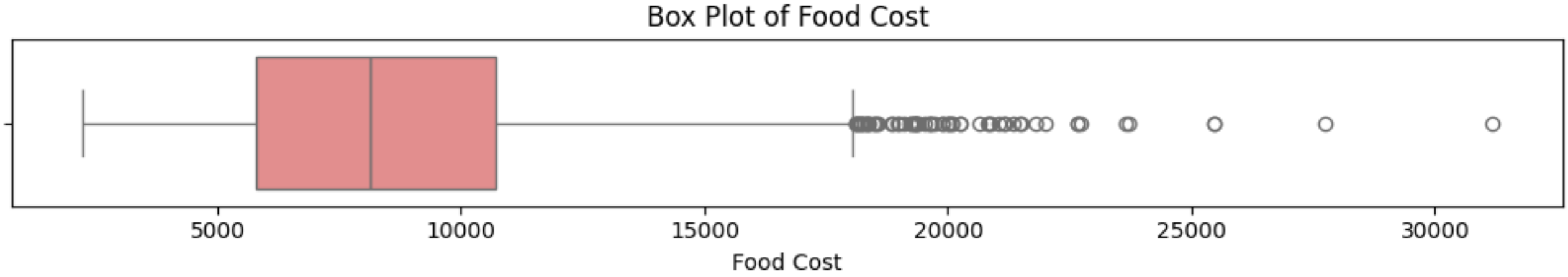


Food Cost

- Range: $28,958.34
  - Min: $2,220.28
  - Max: $31,178.62

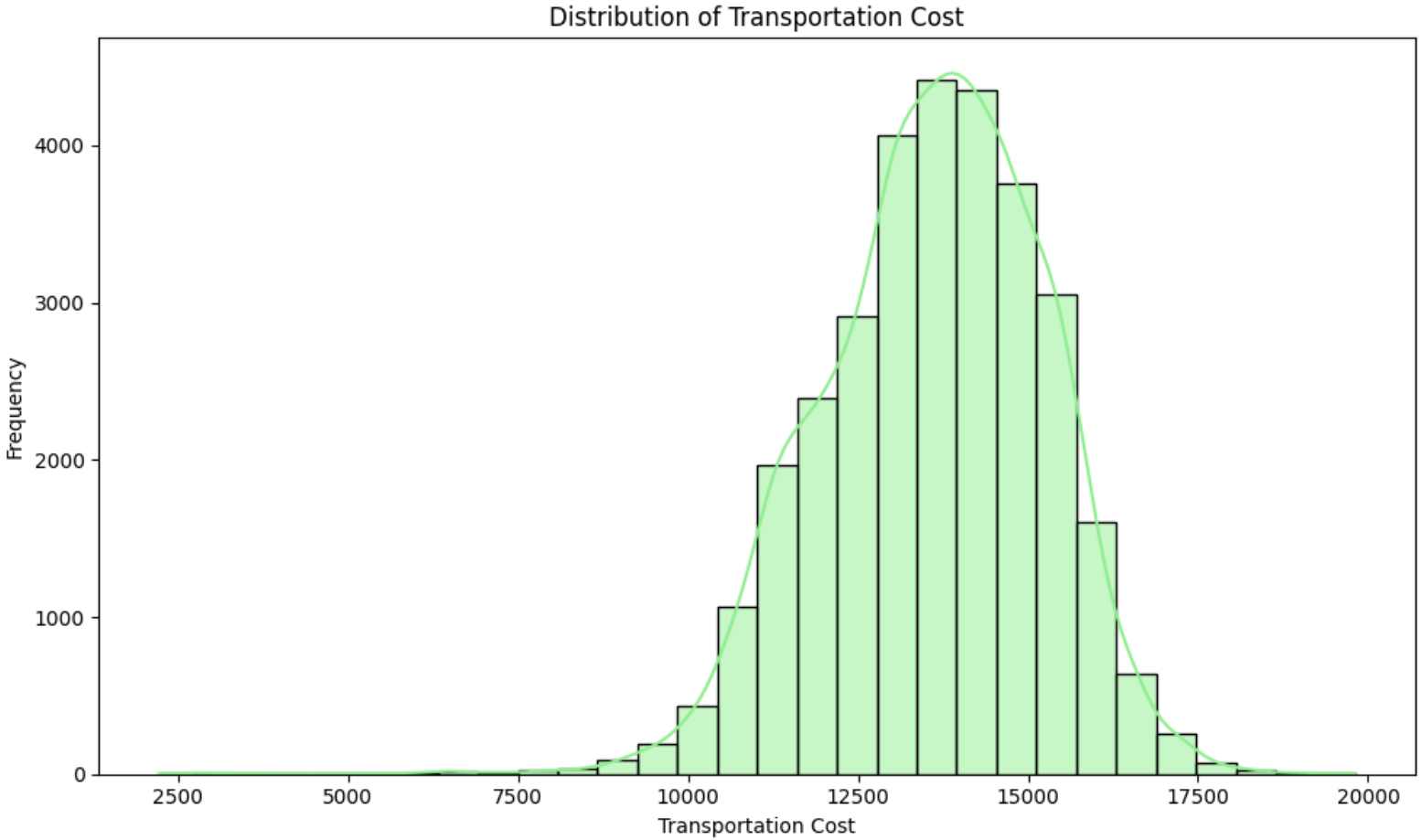
- Average: $8,287.25

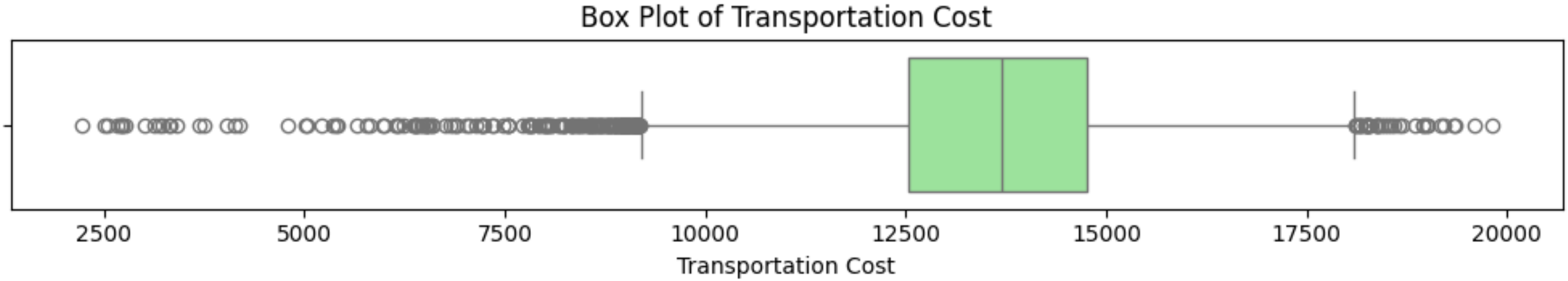




Transportation Cost

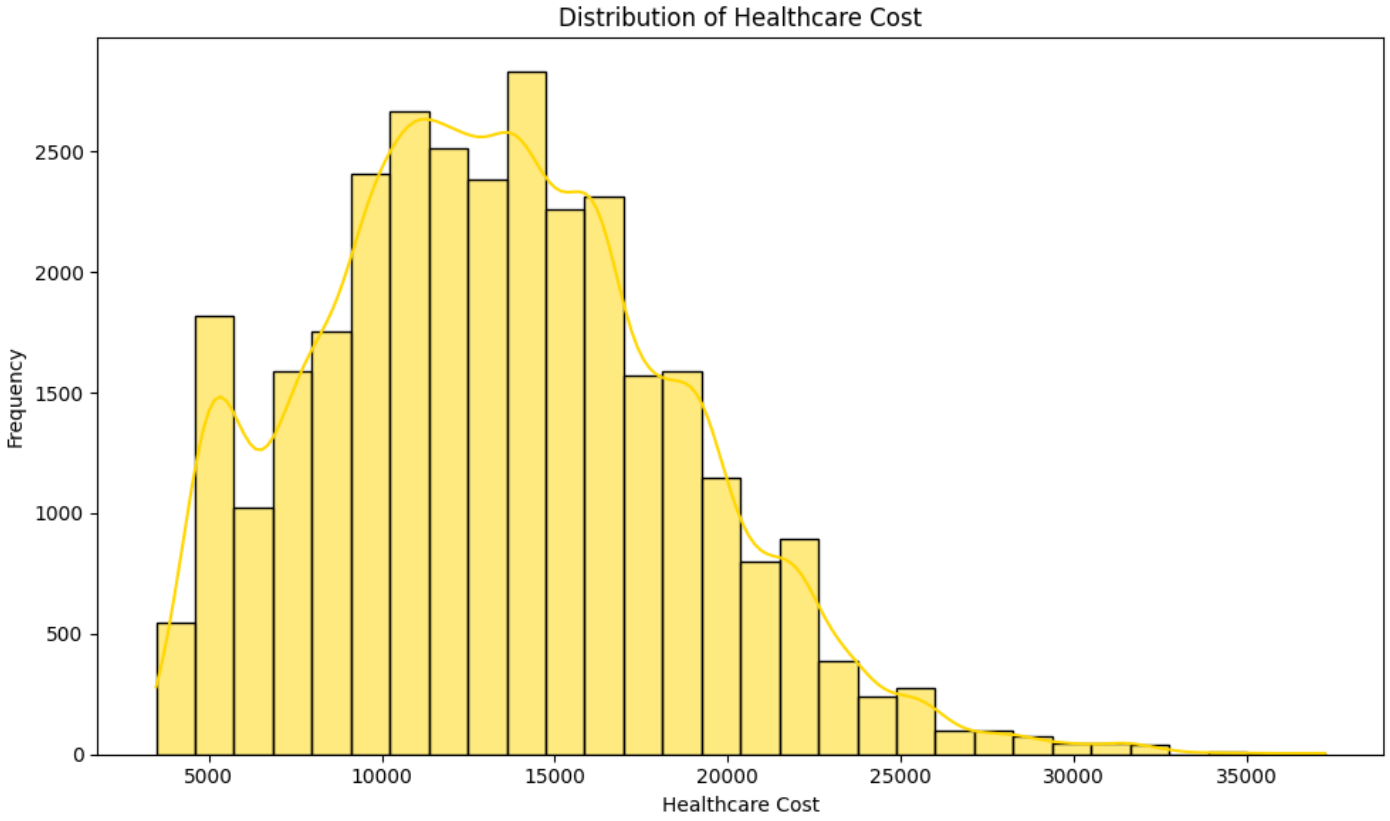
- Range: $17,600.02
  - Min: $2,216.46
  - Max: $19,816.48
- Average: $13,402.99

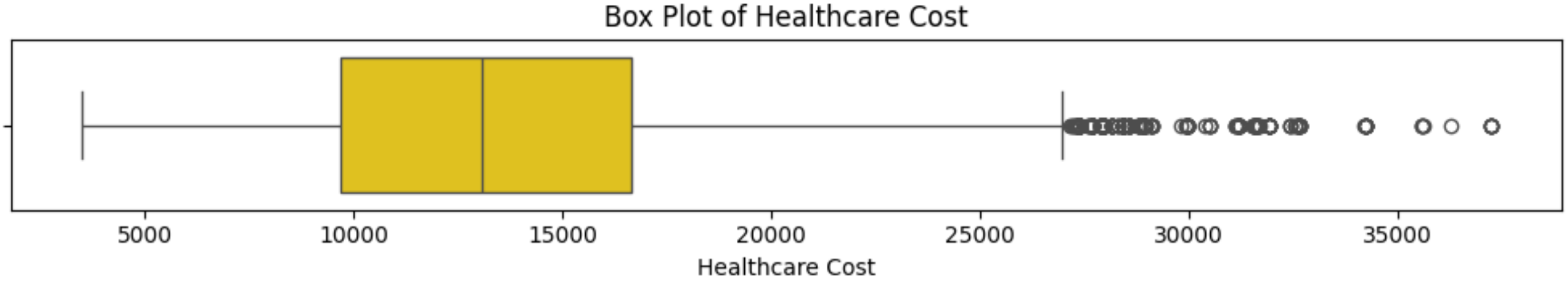




Healthcare Cost

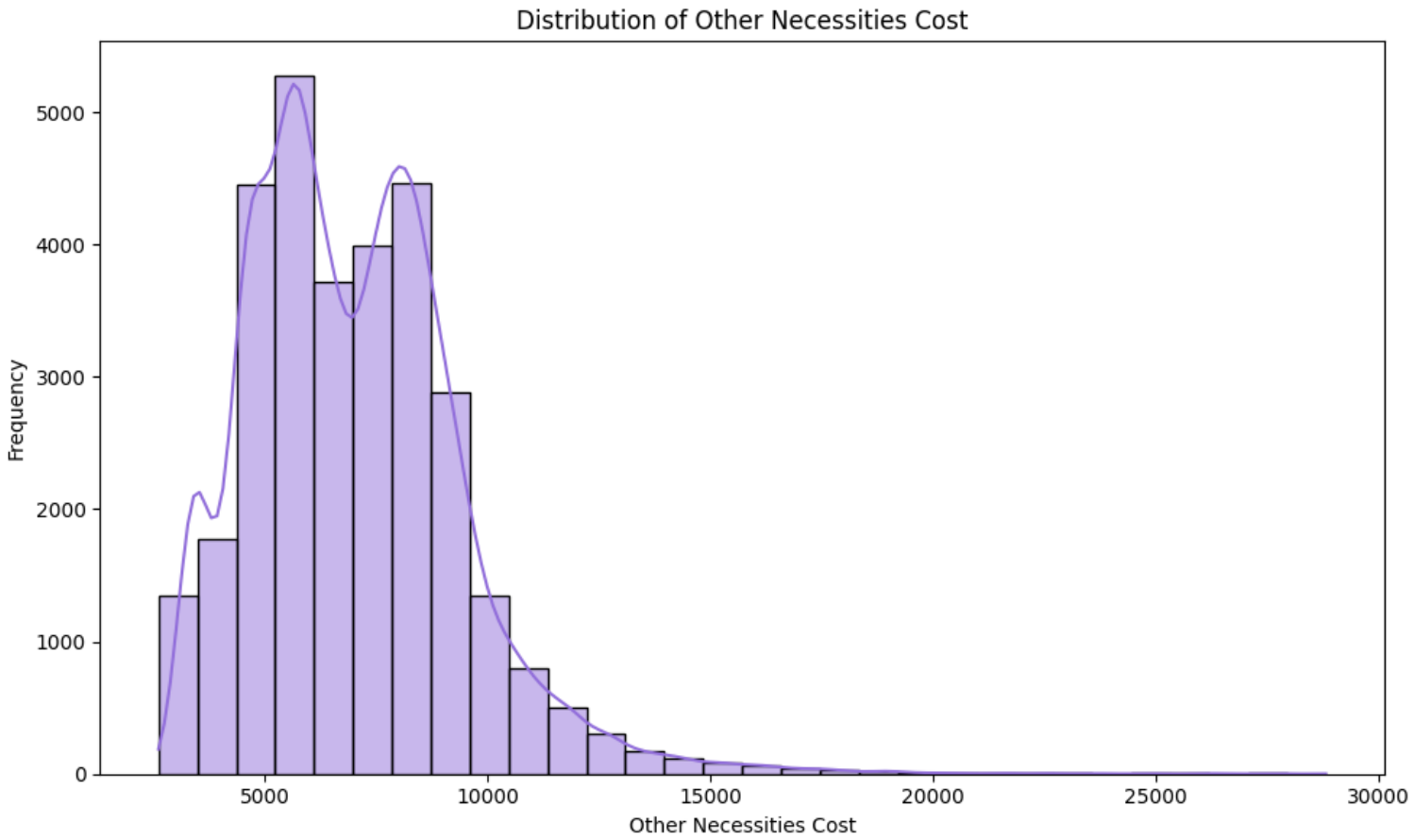
- Range: $33,775.89
  - Min: $3,476.38
  - Max: $37,252.27
- Average: $13,393.01

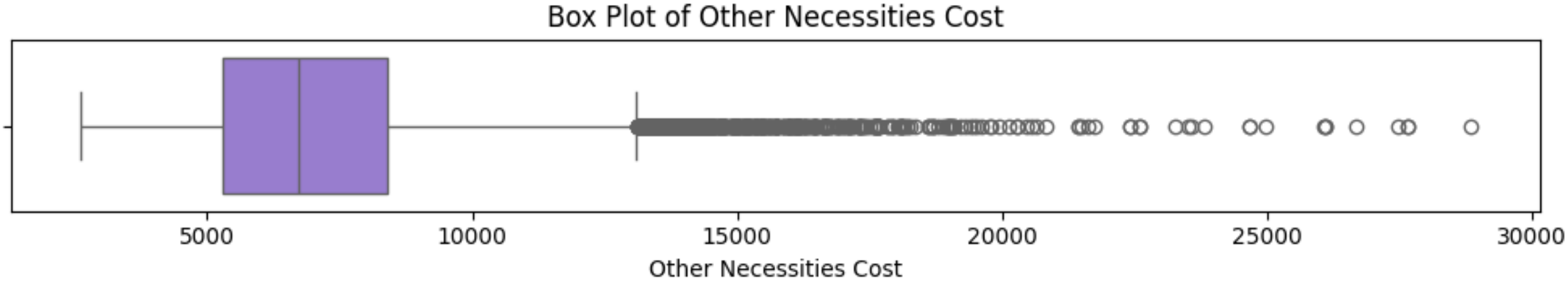




Other

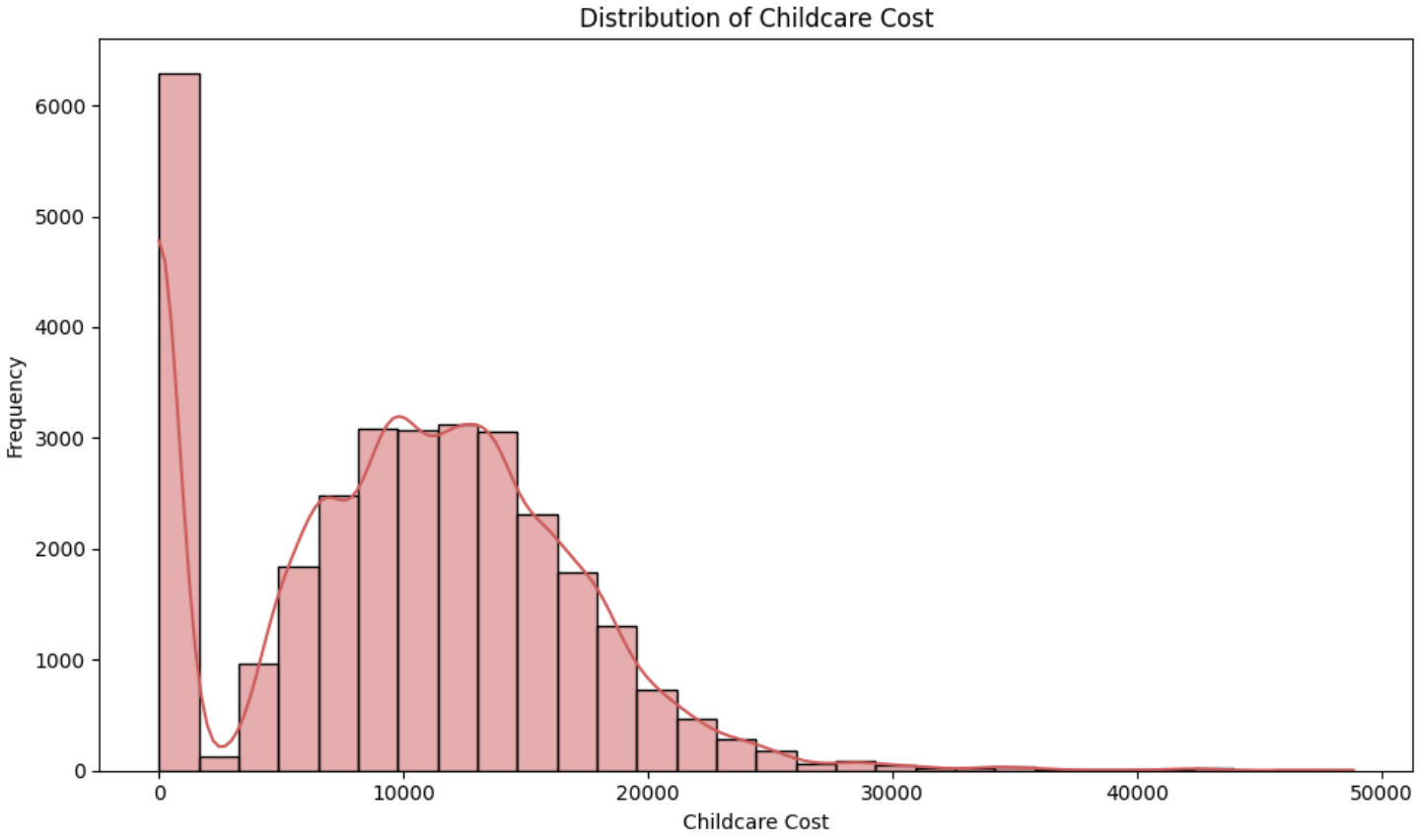
- Range: $26,217.80
  - Min: $2,611.64
  - Max: $28,829.44
- Average: $7,015.26

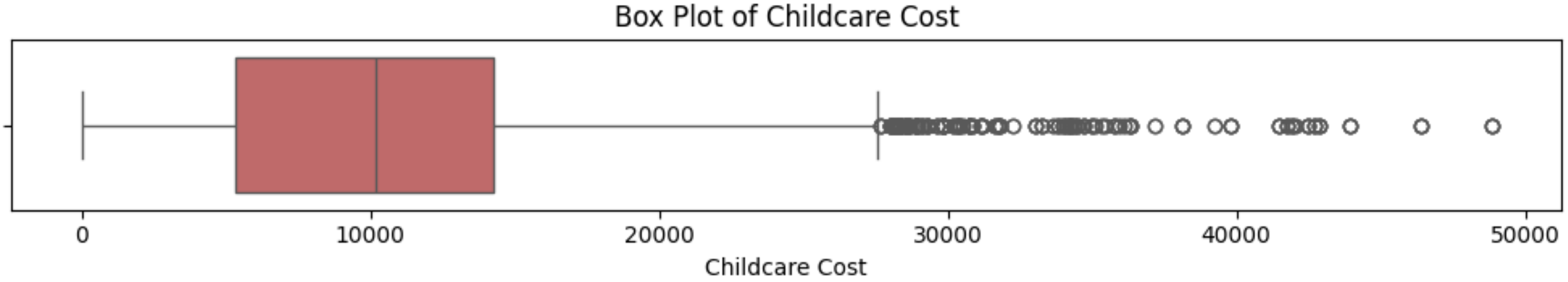




Childcare Cost

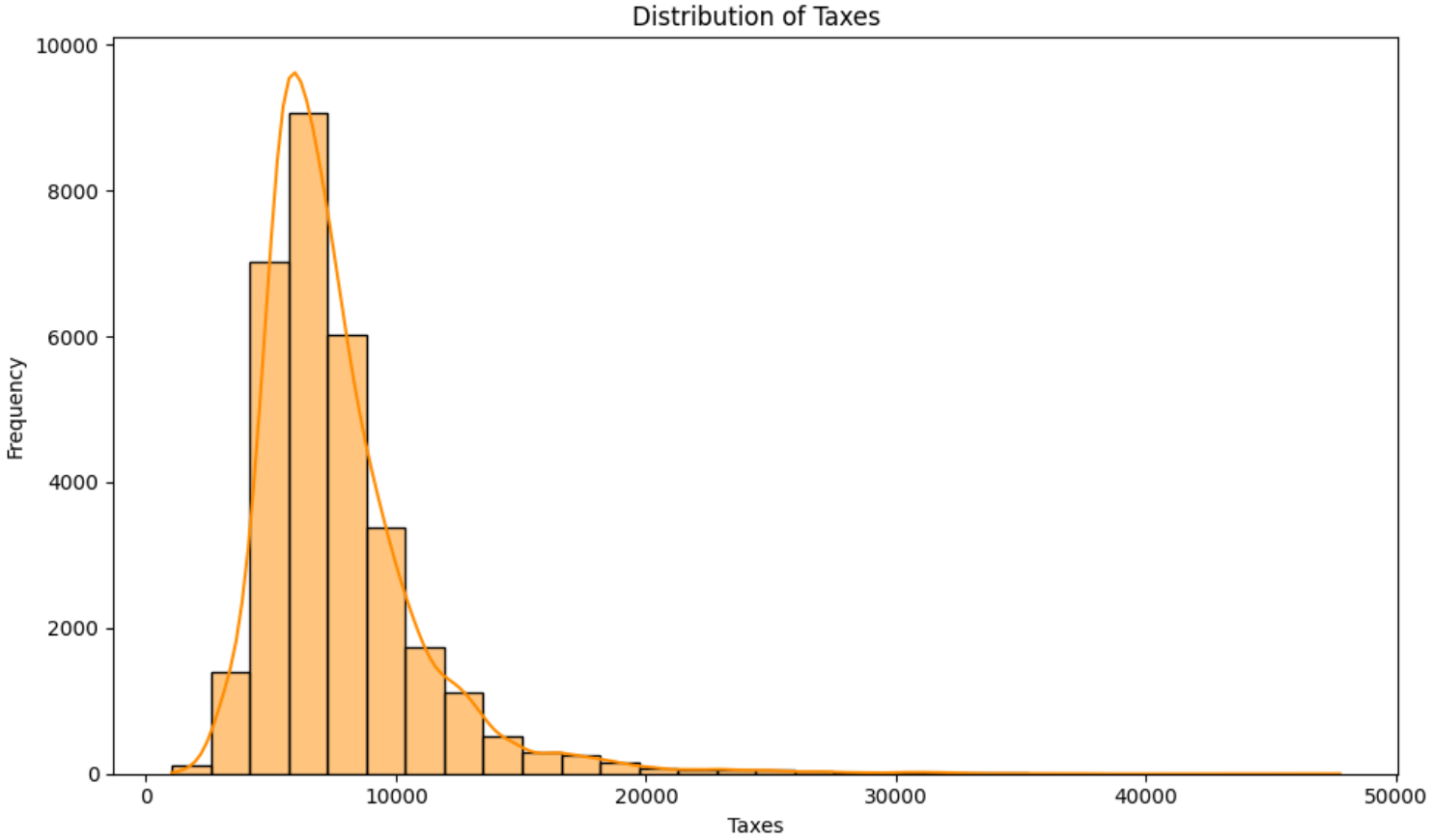
- Range: $48,831.09
  - Min: 0.00
  - Max: $48,310.09
- Average: $9,879.25

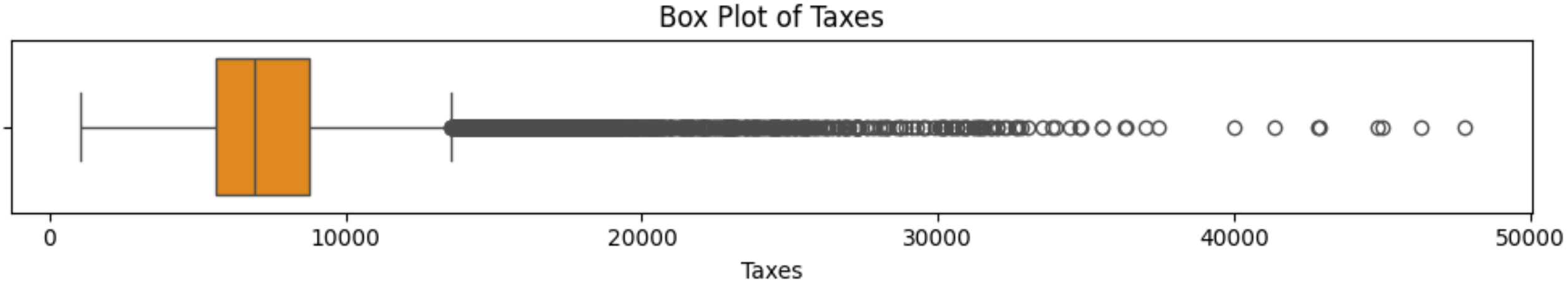




Taxes

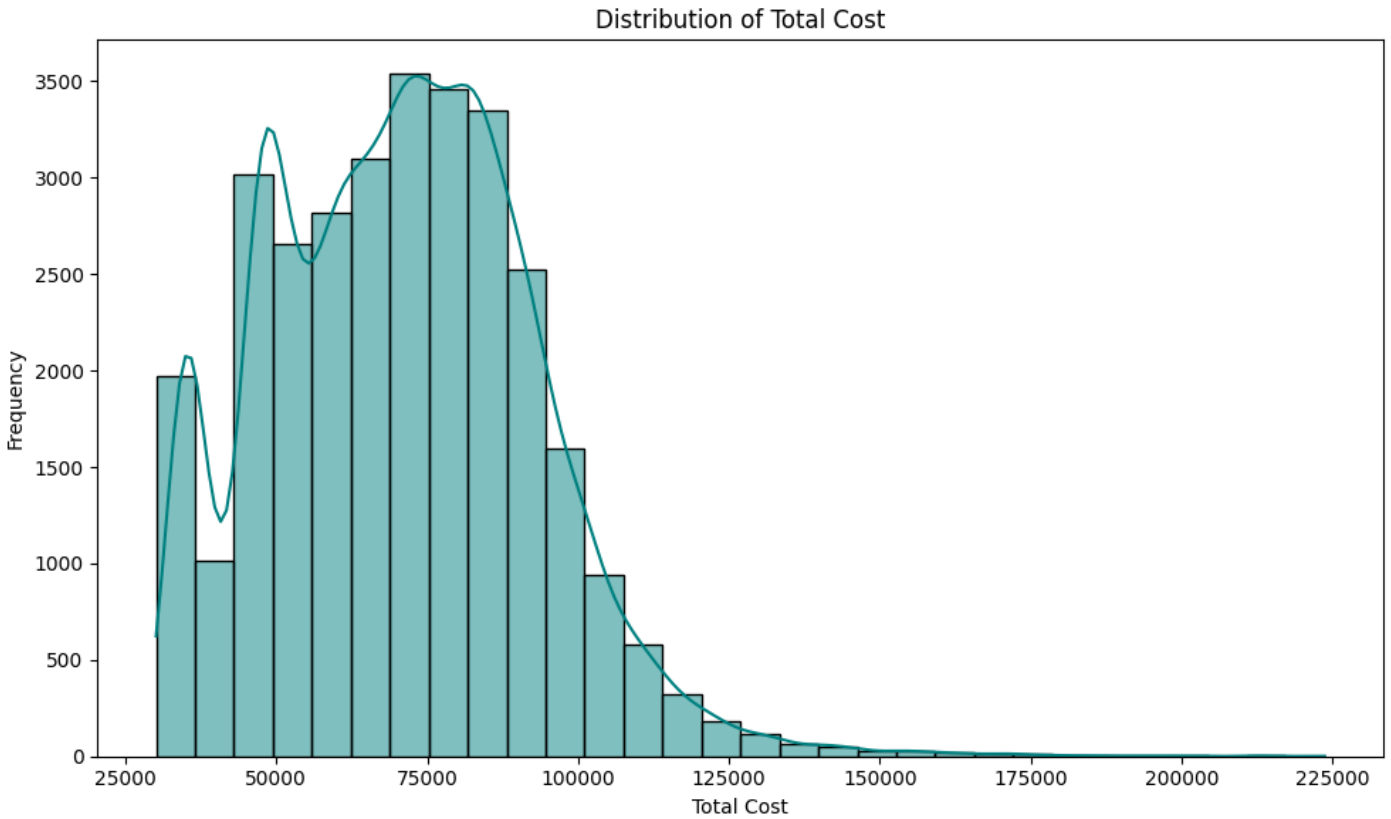
- Range: $46,725.59
  - Min: $1,027
  - Max: $47,753.39
- Average: $7,657.10

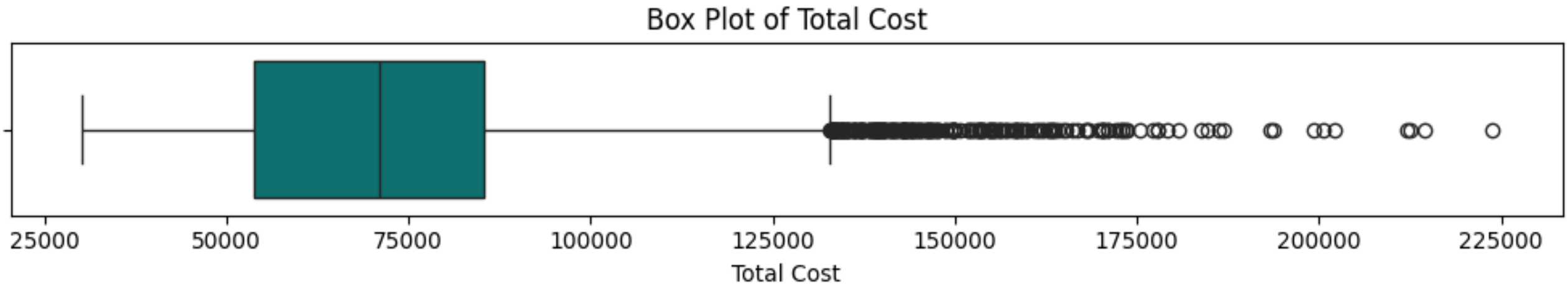




Total Cost

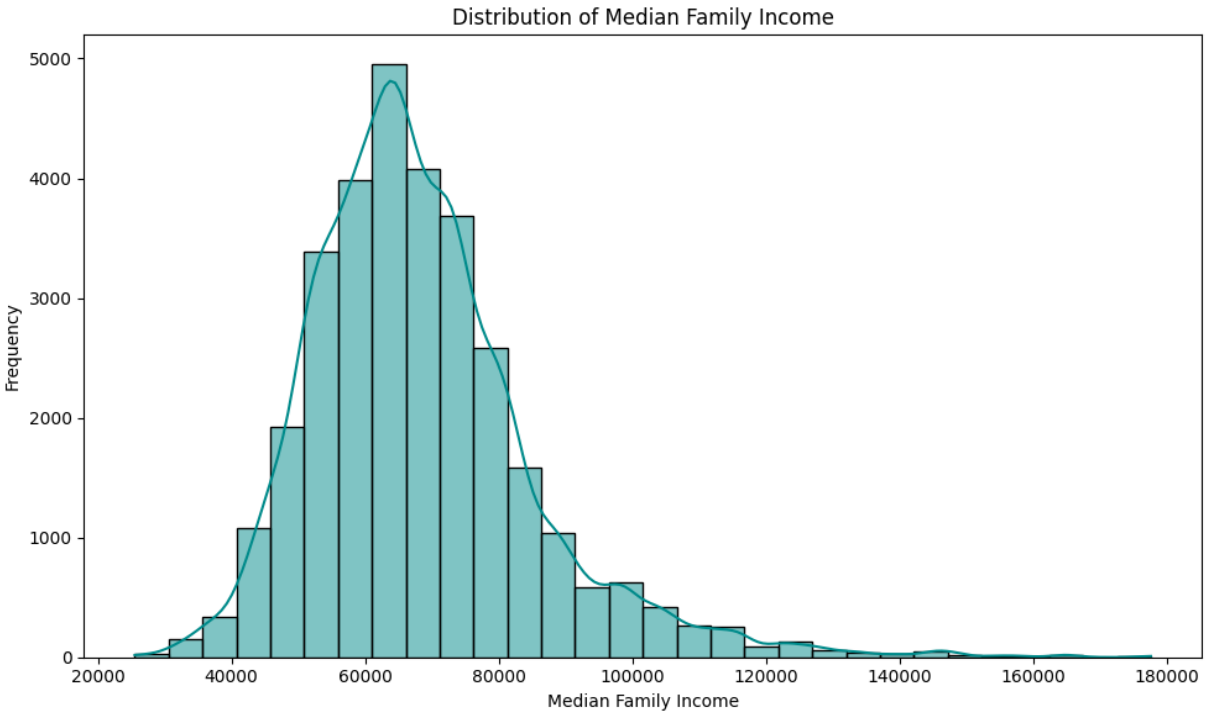
- Range: $193,629.89
  - Min: $30,087.66
  - Max: $223,717.55
- Average: $70,899.62

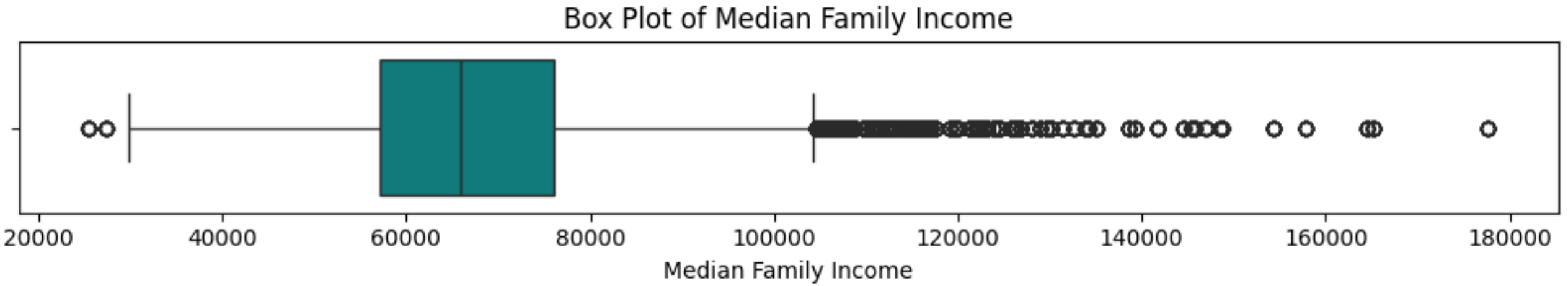




Median Family income

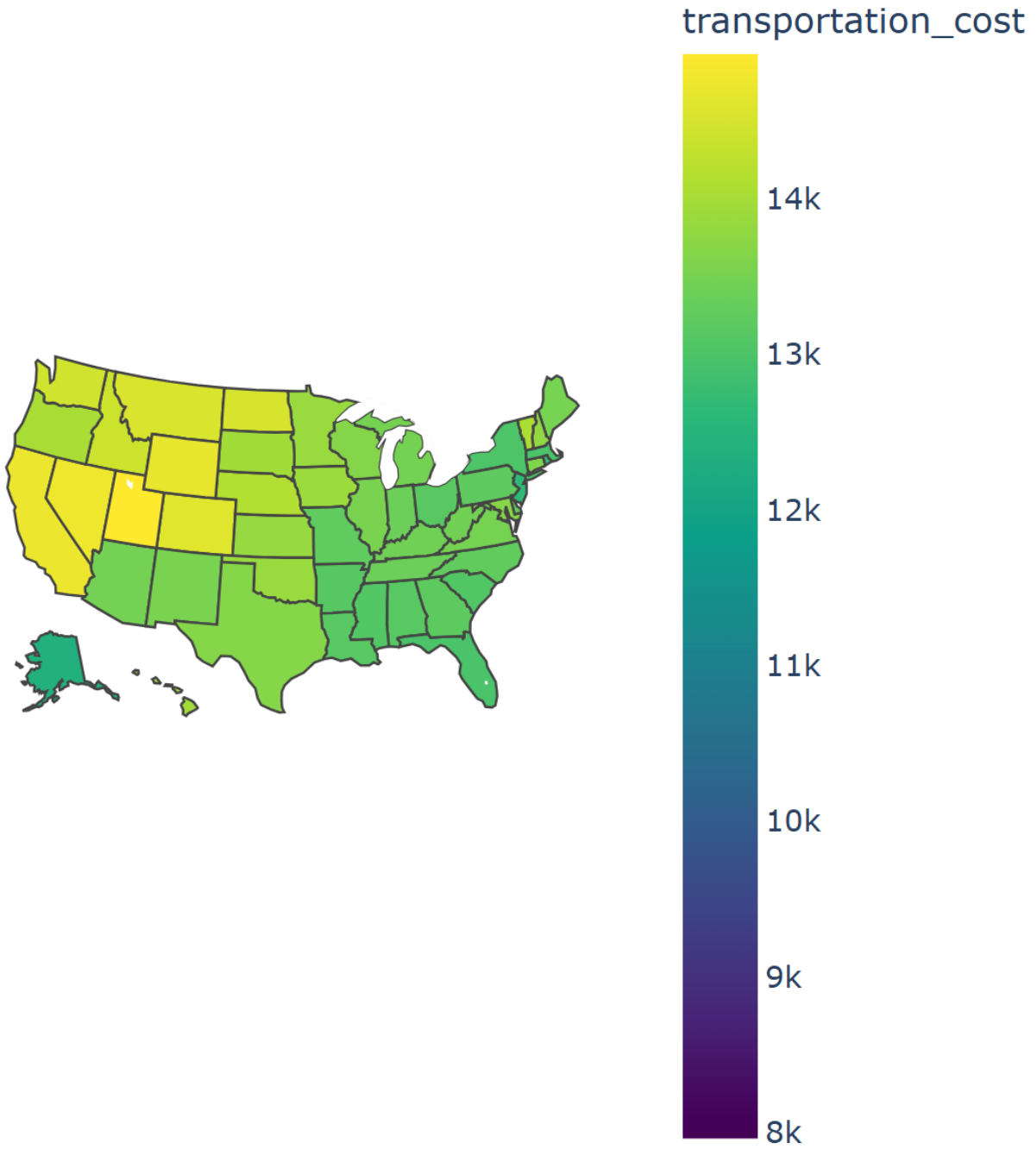
- Range: $68,316.00
  - Min: $25,529.98
  - Max: $177,662.47
- Average: $152,132.49



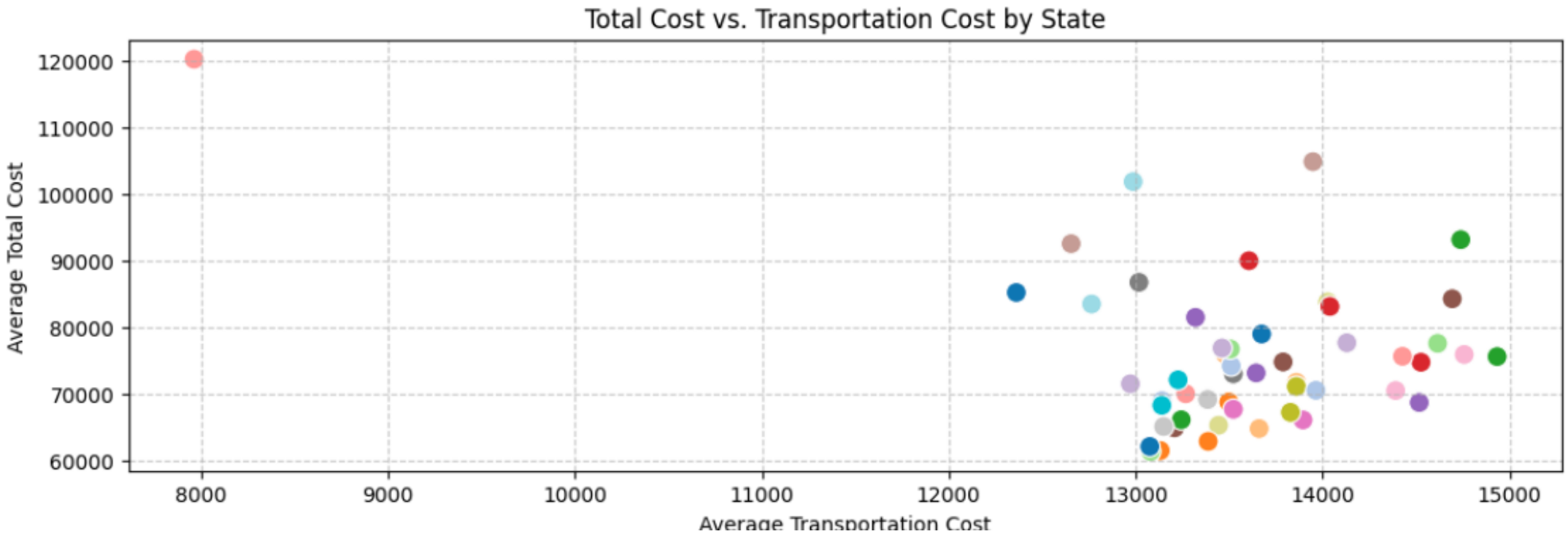


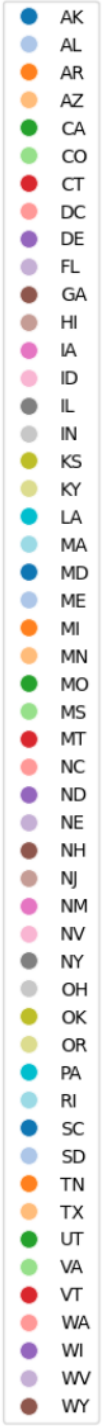
Average Transportation Cost by State

- The western side of the US has higher costs.
- The state with the most Utah
- The state with the lowest is DC

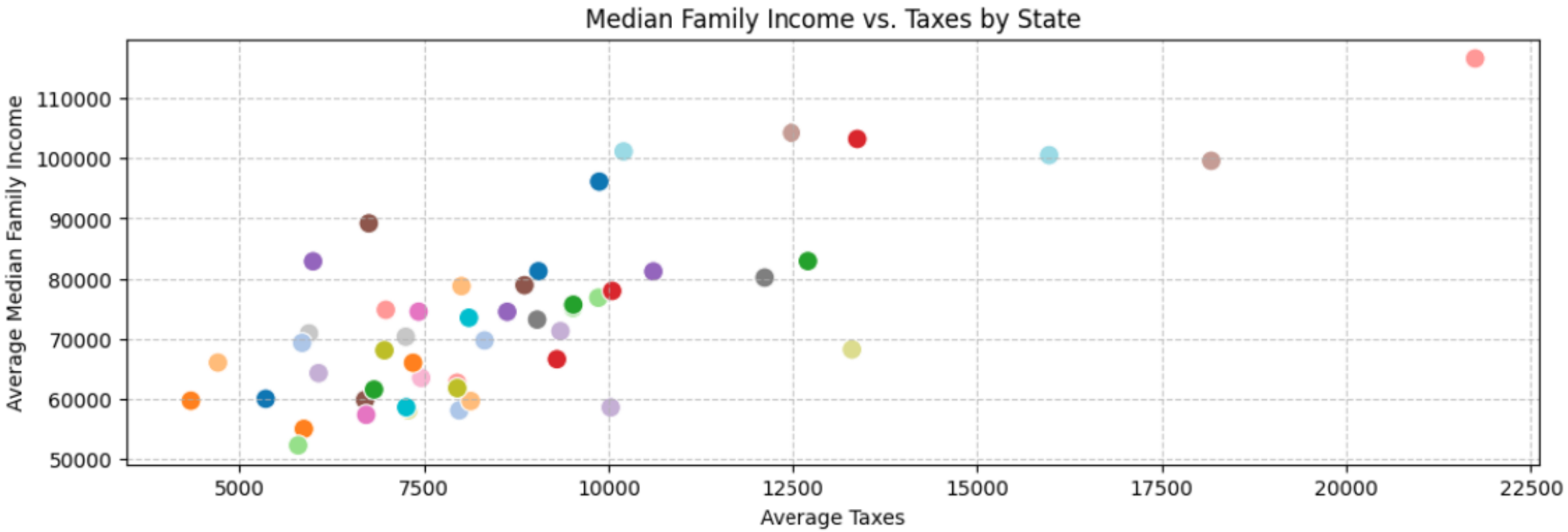


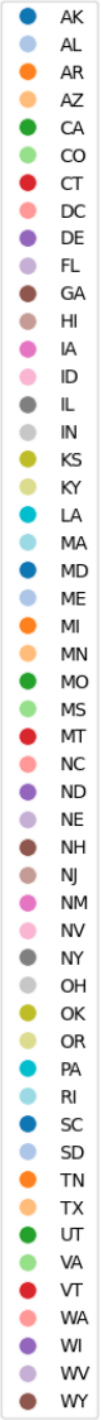
Cost VS Transportation Cost by State



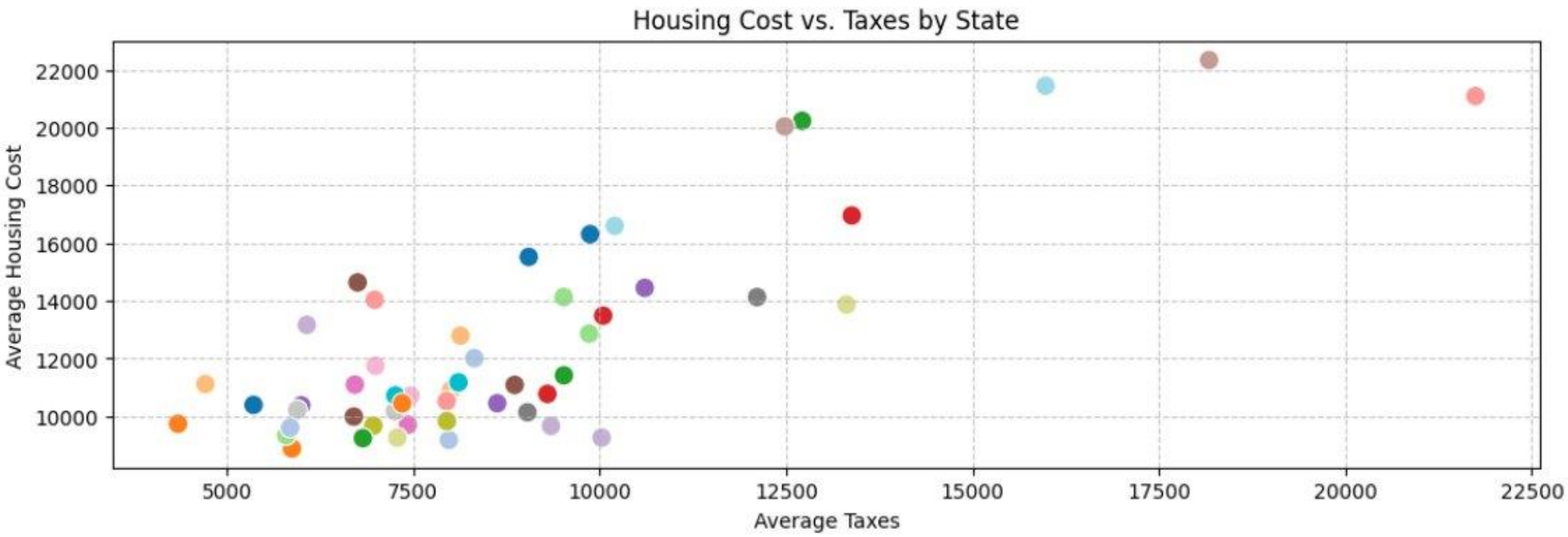


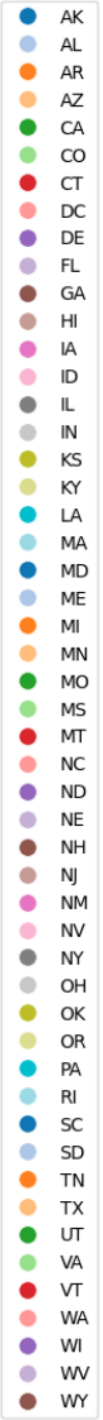
Median Family Income VS Taxes by State



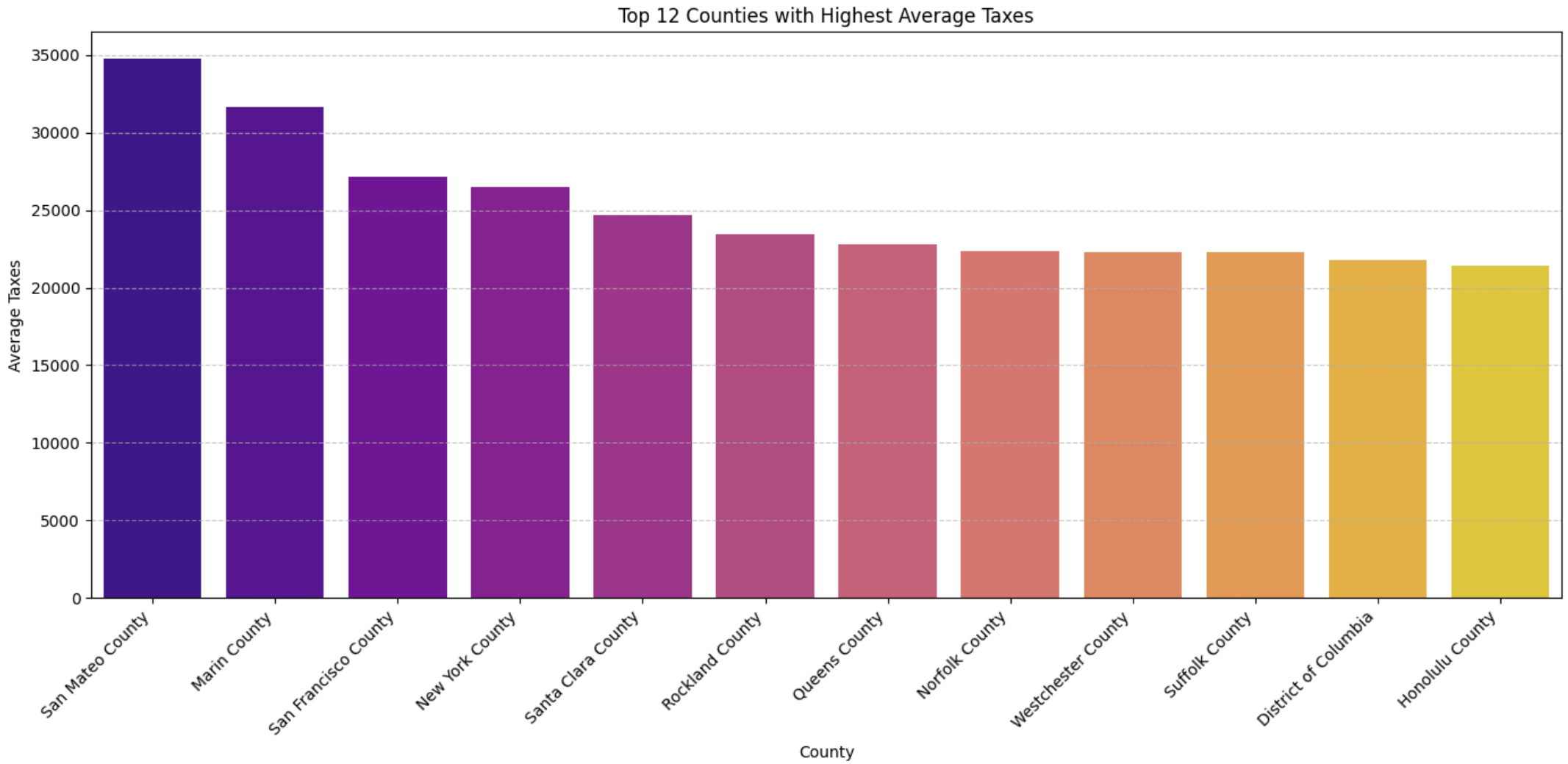


Housing Cost VS Taxes by State

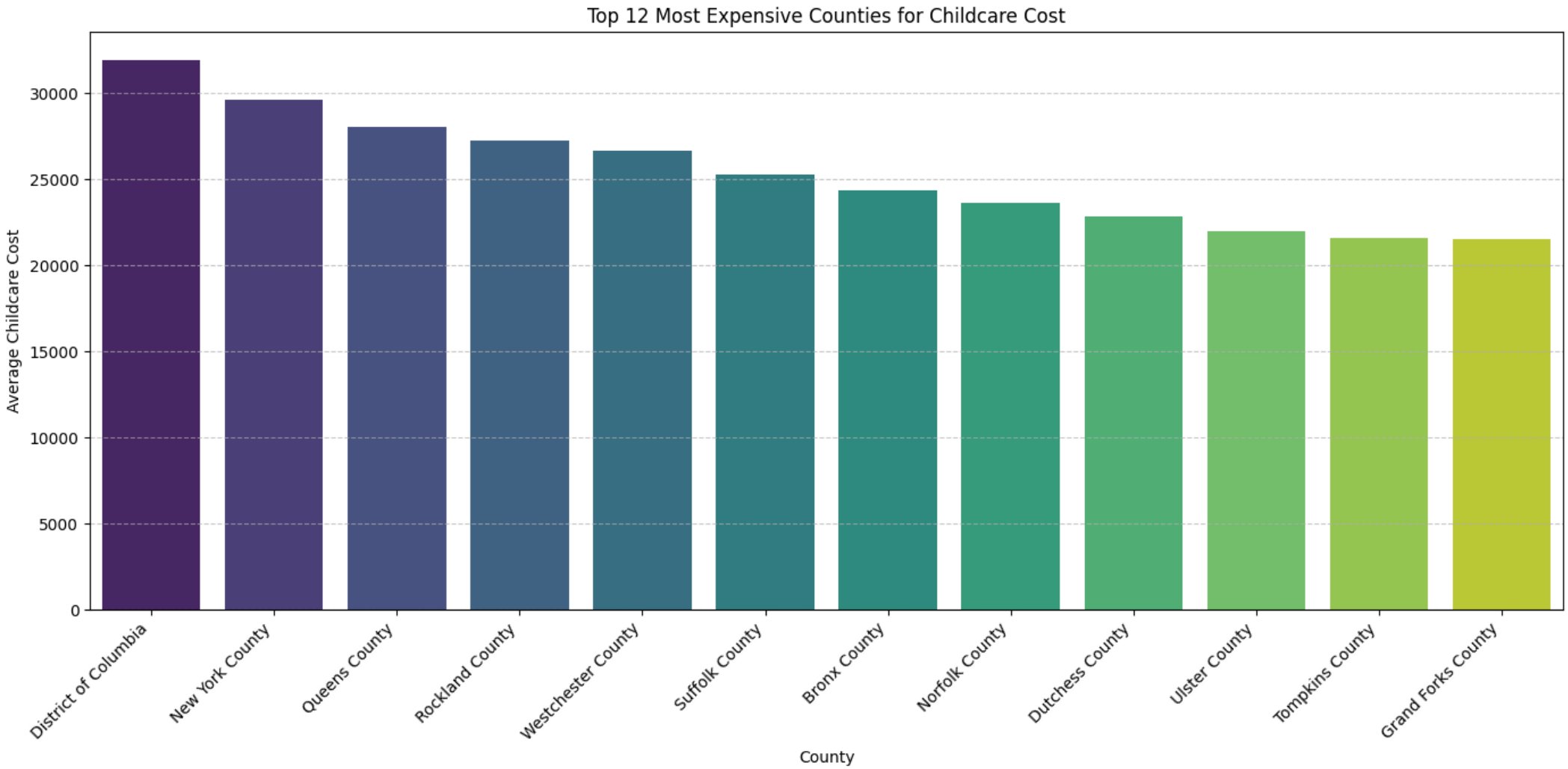


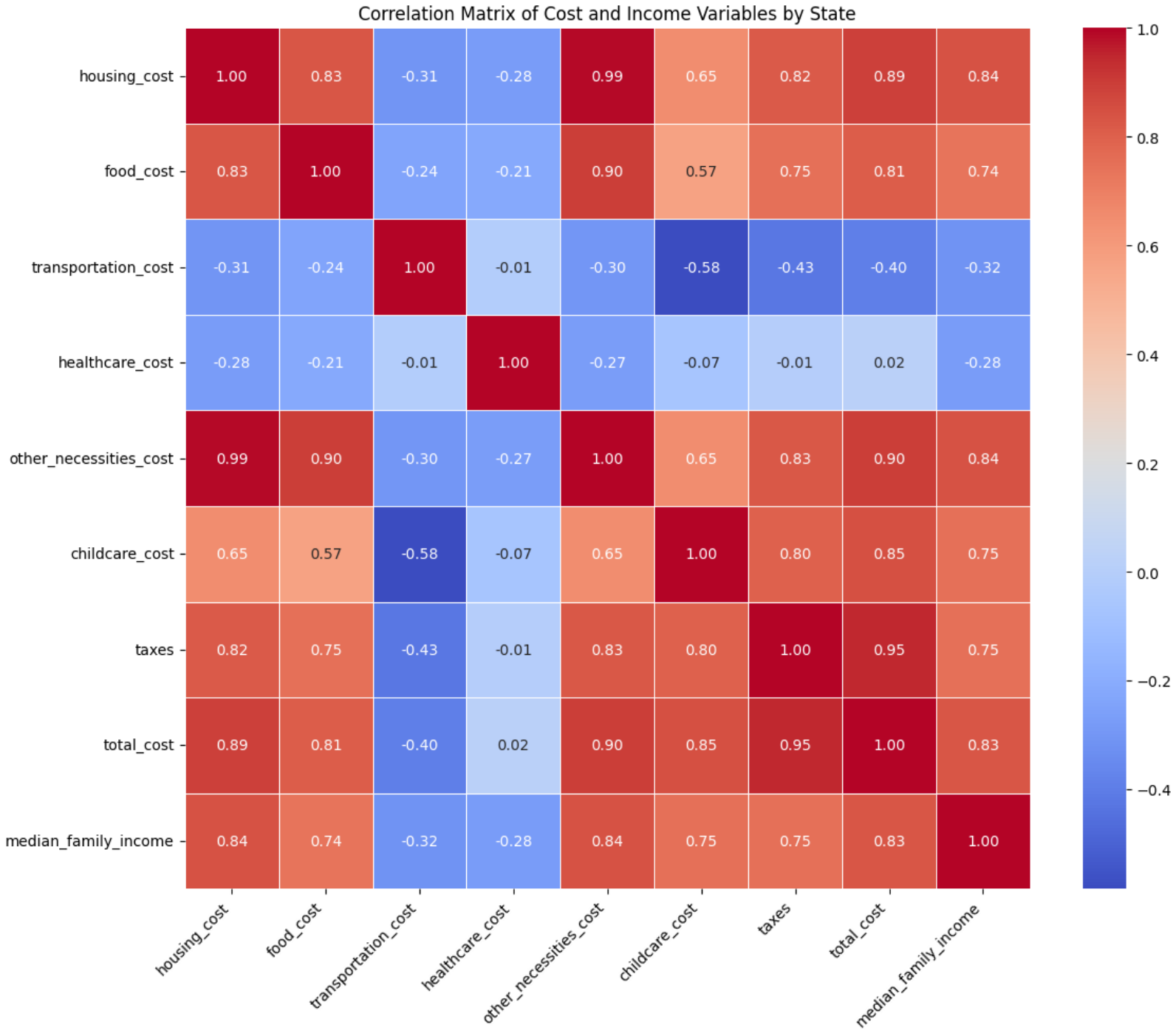


12 Counties with Highest Average Taxes

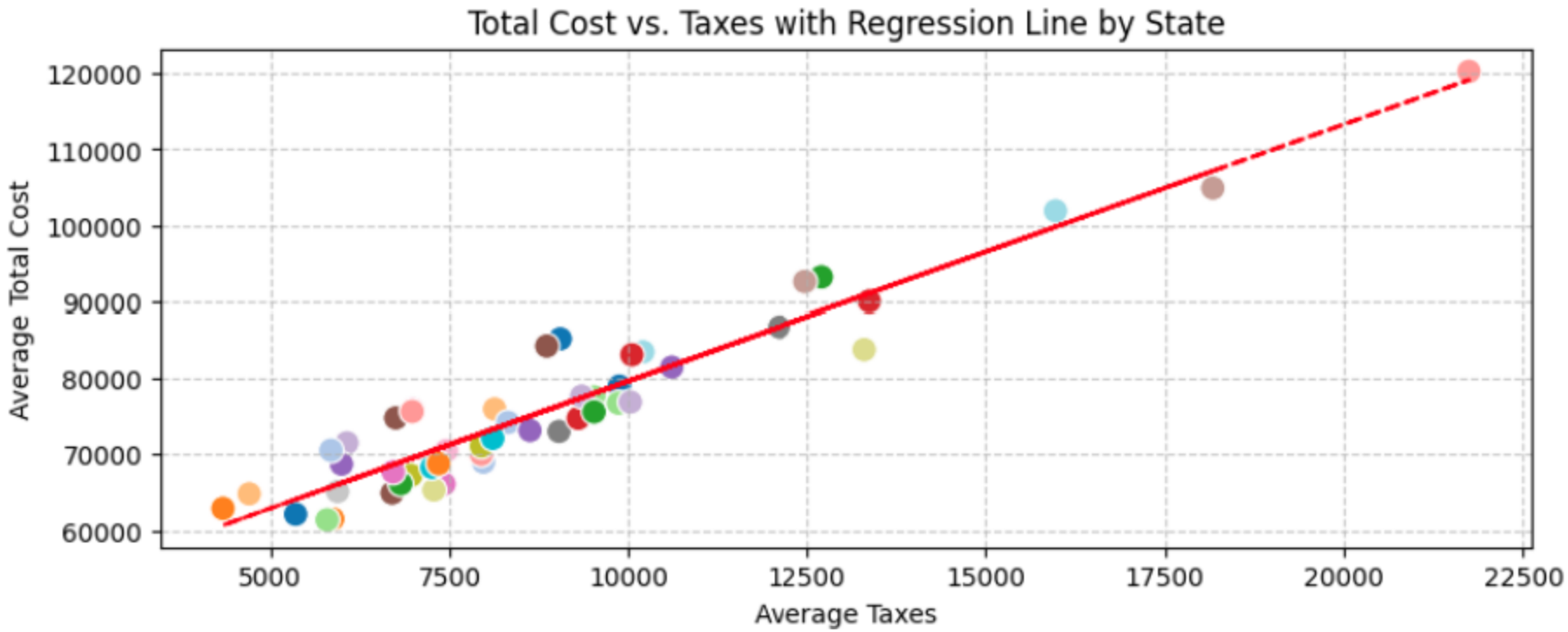


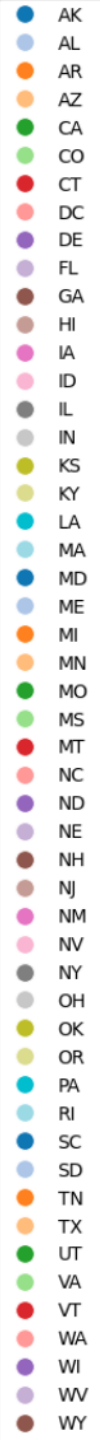
12 Most Expensive Counties for Childcare Cost



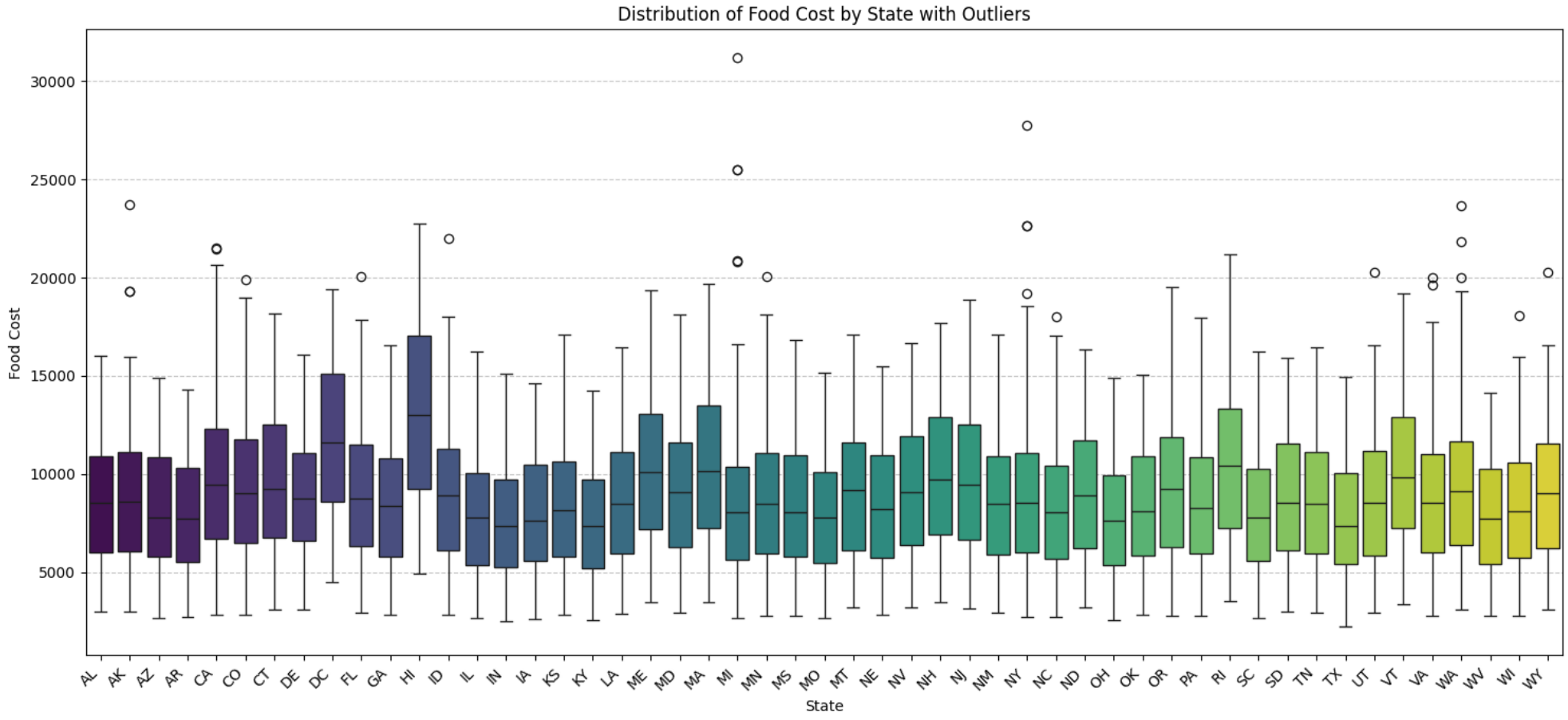

Correlation Matrix of Cost and Income Variables by State
housing_cost
food_cost
transportation_cost
healthcare_cost
other_necessities_cost
childcare_cost
taxes
total_cost
median_family_income
1.00 0.83 -0.31 -0.28 0.99 0.65 0.82 0.89 0.84
0.83 1.00 -0.24 -0.21 0.90 0.57 0.75 0.81 0.74
-0.31 -0.24 1.00 -0.01 -0.30 -0.58 -0.43 -0.40 -0.32
-0.28 -0.21 -0.01 1.00 -0.27 -0.07 -0.01 0.02 -0.28
0.99 0.90 -0.30 -0.27 1.00 0.65 0.83 0.90 0.84
0.65 0.57 -0.58 -0.07 0.65 1.00 0.80 0.85 0.75
0.82 0.75 -0.43 -0.01 0.83 0.80 1.00 0.95 0.75
0.89 0.81 -0.40 0.02 0.90 0.85 0.95 1.00 0.83
0.84 0.74 -0.32 -0.28 0.84 0.75 0.75 0.83 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Total Cost VS Taxes

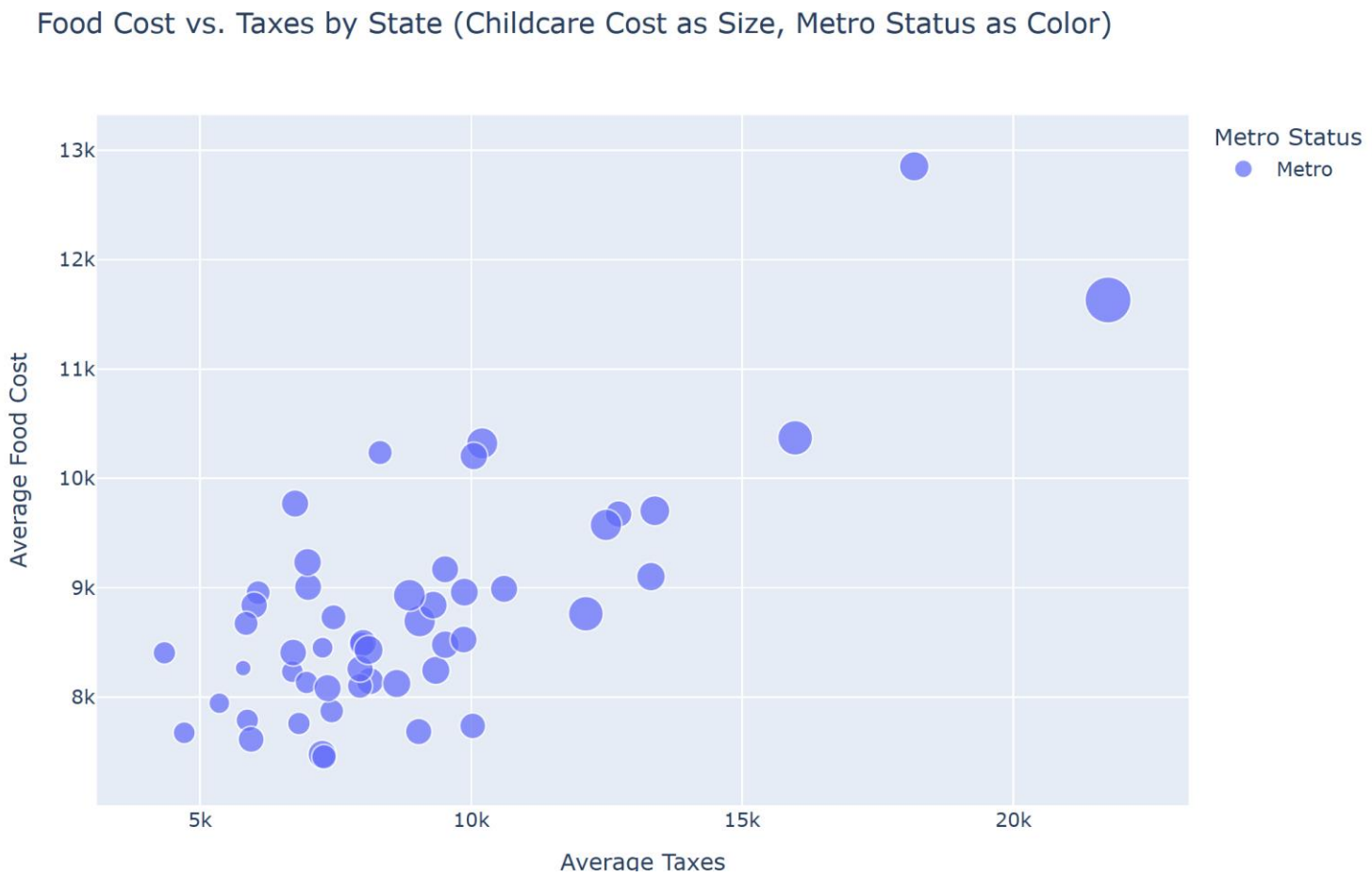




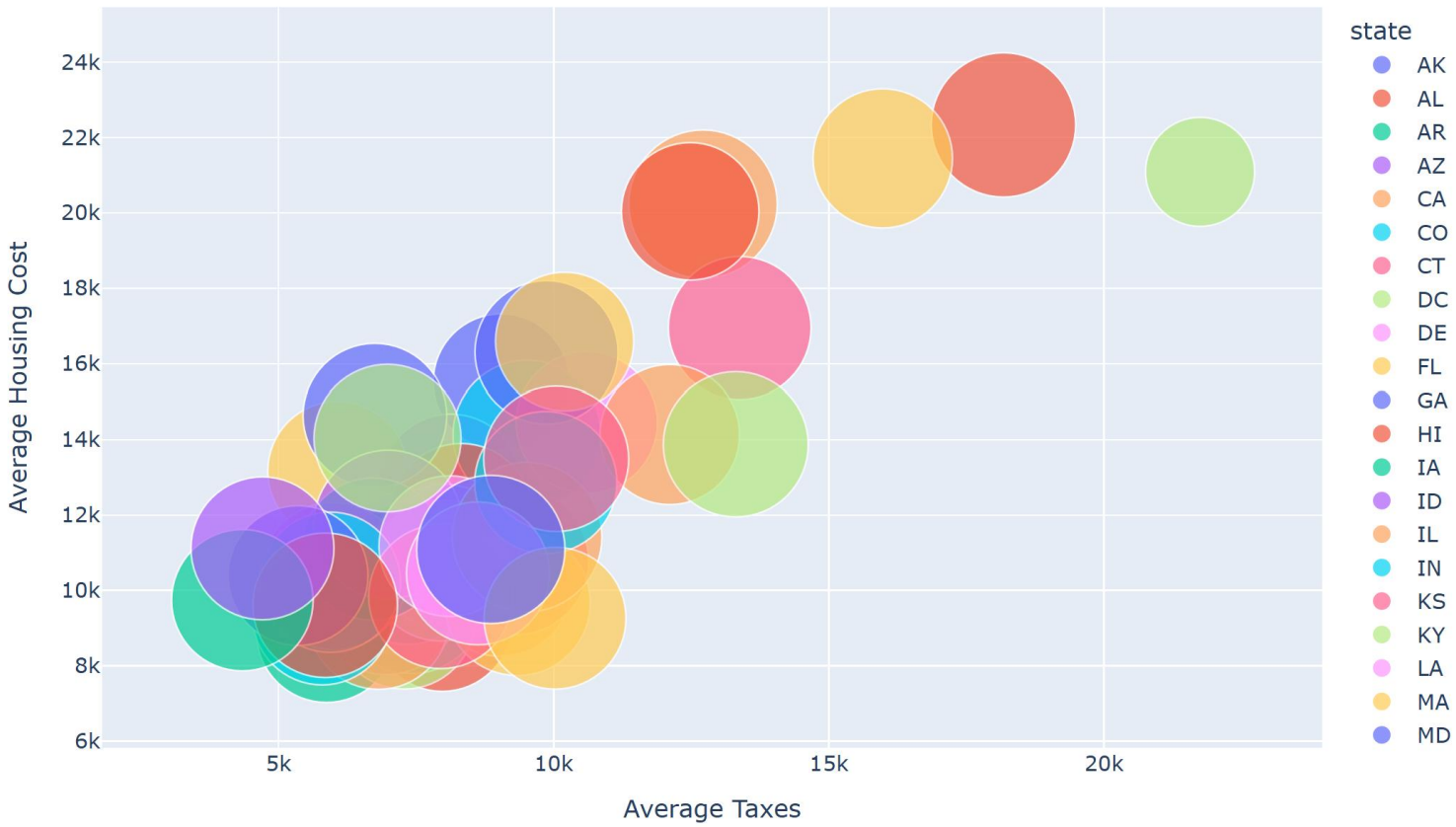
Distribution of Food Cost by State with Outliers



Food Cost VS Taxes by State

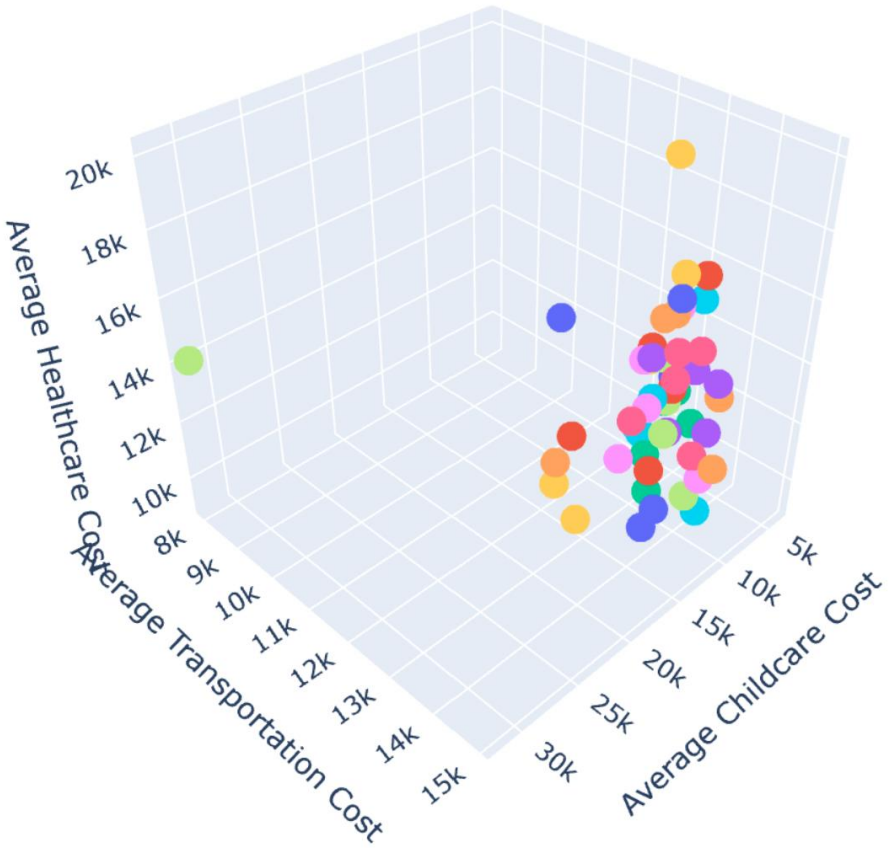


Housing Cost VS Taxes by State

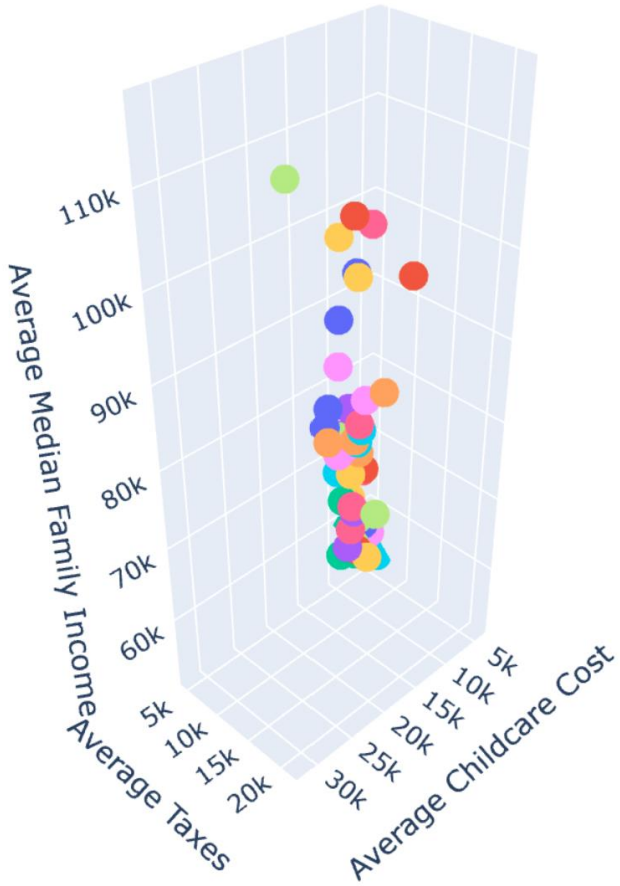


Average Healthcare VS Average Transportation VS Average Childcare

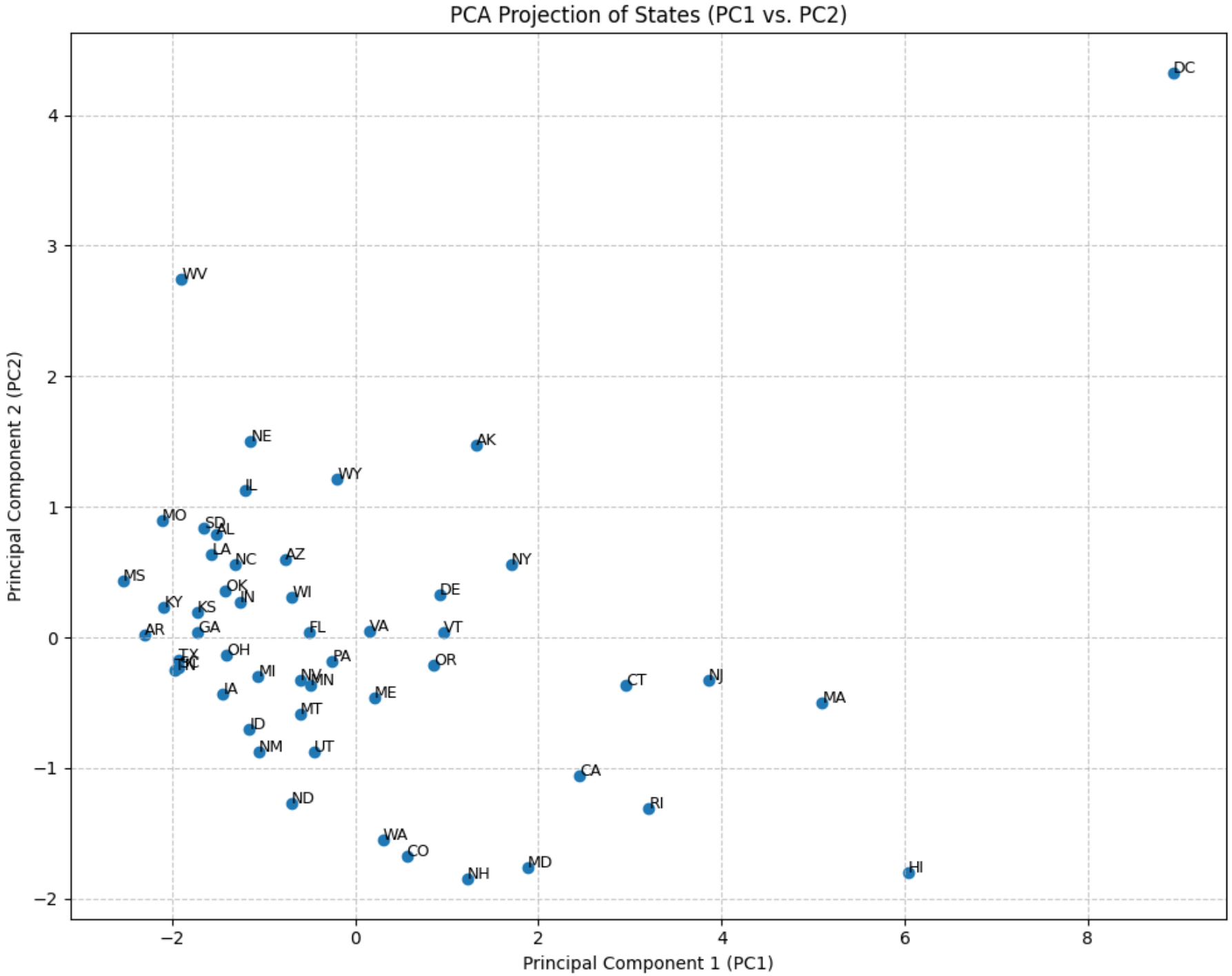
- Lone green dot is Washington DC



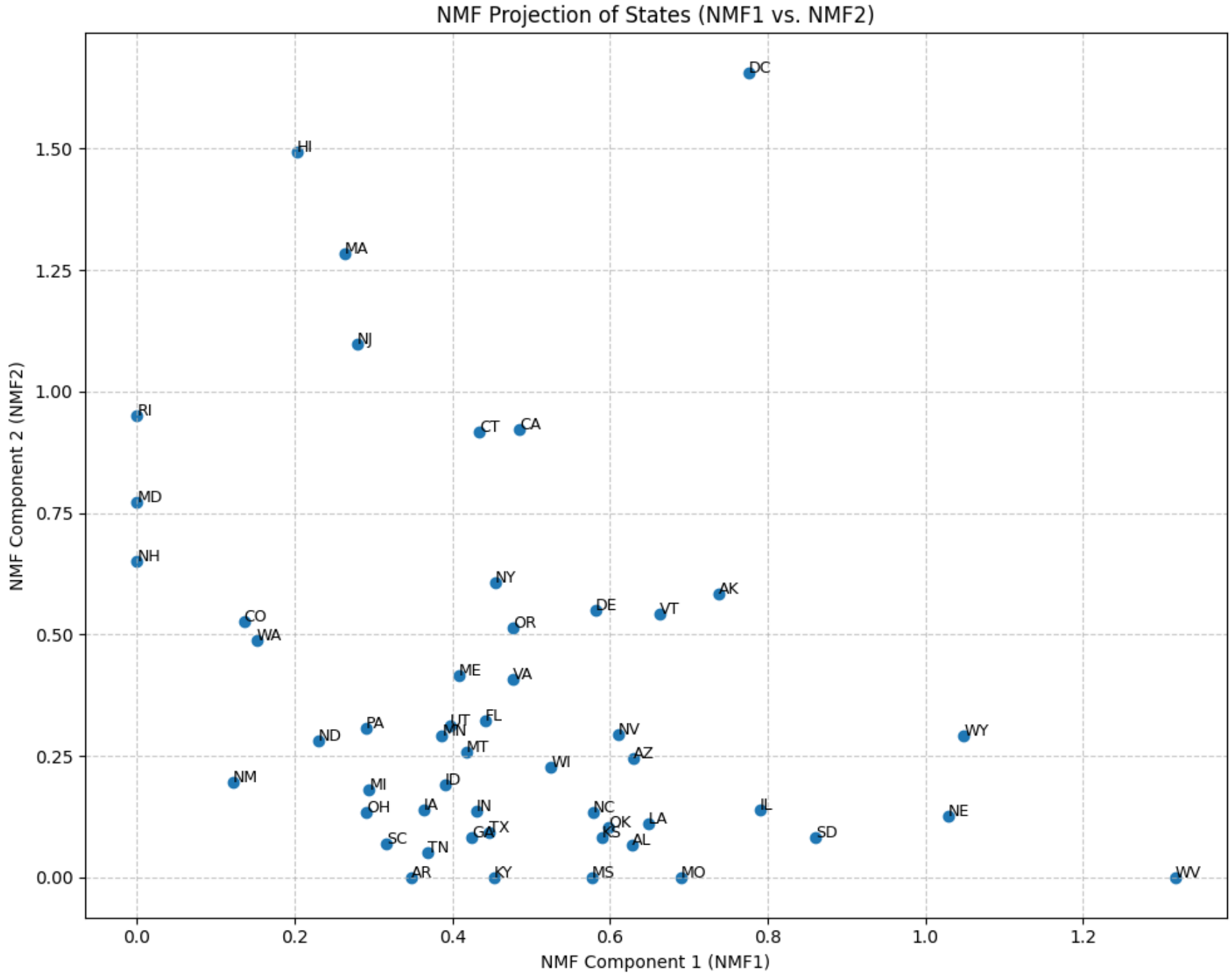
Average Median Family Income VS Average Taxes VS Average Childcare Cost



PCA


PCA Projection of States (PC1 vs. PC2)
Principal Component 2 (PC2)
Principal Component 1 (PC1)
4
3
2
1
0
−1
−2
−2
0
2
4
6
8
DC
WV
NE
AK
WY
IL
MO
SD
AL
LA
NC
AZ
NY
MS
OK
IN
WI
DE
KY
KS
AR
GA
FL
VA
VT
OH
PA
OR
TX
MI
NV
MN
CT
NJ
IA
ME
MA
MT
ID
NM
UT
CA
ND
RI
WA
CO
MD
NH
HI

NMF


NMF Projection of States (NMF1 vs. NMF2)
NMF Component 2 (NMF2)
NMF Component 1 (NMF1)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
0.0
0.2
0.4
0.6
0.8
1.0
1.2
DC
HI
MA
NJ
RI
CT
CA
MD
NH
NY
AK
DE
VT
CO
OR
WA
ME
VA
PA
UT
FL
ND
MN
NV
WY
MT
AZ
WI
NM
MI
ID
OH
IA
IN
NC
IL
NE
OK
LA
TX
GA
KS
SD
SC
AL
TN
AR
KY
MS
MO
WV

New Findings

- Taxes heavily impact the prices of the other costs
- The western side of the US has higher transportation costs
  - Highest transportation cost: Utah $14934.51
  - Lowest transportation cost: DC $7961.97
- Hawaii averages the highest food costs.